

NEW TRENDS AND FRONTIERS IN COMPUTER SCIENCES AND ENGINEERING



All Sciences Academy

*NEW TRENDS
AND FRONTIERS
IN COMPUTER
SCIENCES*

Editor

Prof. Dr. AYŞEGÜL ALAYBEYOĞLU





New Trends and Frontiers In Computer Sciences

Editor: Prof. Dr. Ayşegül ALAYBEYOĞLU

Design: All Sciences Academy Design

Published Date: October 2024

Publisher's Certification Number: 72273

ISBN: 978-625-6314-43-6

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Derin Öğrenme Yaklaşımlarıyla Elektrik Enerjisi Üretim Tahminlerinin Karşılaştırılması^{1*}

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^{1*} Bu çalışma yazarın Muş Alparslan Üniversitesi Nükleer Enerji ve Enerji Sistemleri Anabilim dalındaki “Derin öğrenme yöntemleriyle karşılaştırmalı elektrik enerjisi üretim tahminleri (2024)” isimli yüksek lisans tezinden üretilmiştir.

ÖZET

Bu çalışmada, üç farklı ülke için nükleer enerjiden elektrik üretimi analiz edilmiştir. Bu ülkeleri ABD, Fransa ve Japonya olarak sıralayabiliriz. Çalışmada kullanılan veri setleri, Our World in Data kaynağından elde edilmiş olup, her ülkenin nükleer enerji üretim süreçlerine ilişkin uzun vadeli verilere dayanmaktadır. Araştırmanın önemi, nükleer enerjinin elektrik üretimindeki rolünü ve ülkeler arasındaki üretim farklarını ortaya koyarak enerji politikaları ve stratejileri hakkında önemli bilgiler sunmasıdır. Bu tür kapsamlı bir analiz, enerji sektörü için kritik kararların alınmasına yardımcı olabilir ve gelecekteki enerji üretim trendlerinin öngörülmesine katkı sağlayabilir.

Çalışmanın yeniliği, hem istatistiksel hem de derin öğrenme modellerini kullanarak nükleer enerji üretim tahminlerinin yapılmasında yatmaktadır. Kullanılan modeller arasında Naive, SES, Auto.Arima, Holt-Winters, ETS, Thetaf, NNETAR ve MLP gibi yöntemler bulunmaktadır. Bu modellerin performansı, RMSE, MAE ve MAPE gibi metriklerle değerlendirilmiştir. Analiz sonuçları, her ülke için en uygun modelin farklı olduğunu göstermiştir; örneğin, ABD için NNETAR modeli en başarılı sonuçları verirken, Fransa için Thetaf modeli öne çıkmıştır. Bu durum, her ülkenin enerji üretim dinamiklerinin farklı olduğunu ve bu nedenle model seçiminde dikkatli olunması gerektiğini göstermektedir.

Çalışmanın literatüre katkısı, nükleer enerji üretiminin analizi ve tahmininde kullanılan model çeşitliliği ve bu modellerin farklı ülkeler üzerindeki performansının incelenmesiyle belirginleşmektedir. Enerji üretiminde kullanılan yöntemlerin karşılaştırılması, politika yapıcılar ve araştırmacılar için değerli bilgiler sunmaktadır. Ayrıca, bu çalışmada elde edilen bulgular, enerji güvenliği ve sürdürülebilirlik konularında önemli çıkarımlar yapılmasına imkân tanımaktadır.

Genel olarak, bu araştırma, enerji sektöründeki modelleme çalışmalarına önemli bir katkı sağlayarak, gelecekteki enerji üretim trendlerini daha doğru bir şekilde tahmin etmeye yardımcı olabileceğine inanılmaktadır.

Anahtar Kelimeler – Nükleer enerji, elektrik üretimi, zaman serisi analizi, derin öğrenme modelleri, enerji politikası, tahmin modelleri

GİRİŞ

Enerji, modern toplumun hayatını idame ettirebilmesi için en temel gereksinimlerden biridir. Endüstri, ulaşım, iletişim, sağlık, eğitim gibi hayatımızın her alanında enerjinin rolü büyük ve kaçınılmazdır.

Enerji, günümüzde çeşitli kaynaklardan elde edilmektedir. Fosil yakıtlar, nükleer enerji, güneş enerjisi, ısı enerjisi, kinetik enerjisi, kimyasal enerji gibi kaynaklar enerji ihtiyacımızı karşılamada kullanılmaktadır.

2020 yılı itibariyle Dünyada 57 yıllık petrol, 49 yıllık doğal gaz ve 139 yıllık kömür rezervi kalmıştır ve bu sürenin sonunda fosil yakıtlar tükenecektir. Dünya genelinde enerji ihtiyacının stratejik ve çok kritik bir durumda olması nedeniyle, enerjide dışa bağımlı olan ülkemizde yenilenebilir enerji alanında yatırımlar hızla artmaktadır. Ülkemizde enerjide dışa bağımlılıktan kurtulabilmek için Mersin Akkuyu’da Rus tasarımı VVER-1200 reaktörü ile donatılmış 4 güç ünitesi aynı anda inşa edilmektedir, 2023 yılı itibariyle nükleer santrale yakıt çubukları getirilmiştir ve 2024 yılında ilk elektrik üretimi plânlanmıştır. Sinop’ta kurulması planlanan ikinci nükleer enerji santrali çalışmaları devam etmektedir.

Ülkemizde rüzgâr enerjisi santrali, güneş enerjisi santrali, jeotermal enerji santrali, biyogaz enerjisi santrali sayısında sürekli bir artış vardır. Çanakkale, Balıkesir, Samsun, Antakya, Karaman, Afyon illerinde rüzgâr santralleri kurulmaktadır. Ocak 2019 tarihi itibariyle Ülkemizde kurulu güç santralleri içerisinde yenilenebilir enerjinin payı, hidroelektrik enerjisinde %30, rüzgâr enerjisinde %8,40, güneş enerjisinde %1,52, jeotermal enerjide %2,96 seviyesindedir.

Bu çalışmada bilime ve hayatımıza katkı sağlamak adına nükleer enerji üretiminin en çok yapıldığı üç ülkenin elektrik üretim verileri üzerinde tahminlerde bulunduk. Çalışmada istatistiksel ve derin öğrenme modelleri kullanıldı. Bu modeller; NAIVE, SES (Simple exponential Smothing), AUTO.ARIMA (AutoRegressive Integrated Moving Average), HOLT-WINTER, ETS (Exponential Smoothing), THETAF, NNAR (Neural Network Autoregressive), MLP (Multilayer Perceptrons) modelidir.

MATERYAL ve YÖNTEM

Bu çalışmada nükleer enerjiden elektrik enerjisi üretimi ile ilgili üç adet farklı veri seti kullanılmıştır. ABD, Fransa ve Japonya’ya ait veri setleri 58-yıllık verilerden oluşmaktadır. Burada kullanılan tüm veriler Our World in Data'dan (<https://ourworldindata.org/energy>) elde edilmiştir. Buradan tedarik

edilerek kullanılan veri setlerinin tamamı nükleer enerji, üretimine diğer bir ifadeyle elektrik enerjisinin nükleer enerjiden elde edilmesine dayanmaktadır.

Tedarik edilen her bir veri seti kullanılmadan varsa içerdikleri hatalı datalar düzeltilmiş ve daha sonra zaman serisi formatına dönüştürülmüştür.

Zaman serileri, belirli bir zaman dilimi boyunca düzenli aralıklarla toplanan verilerden oluşur ve genellikle trendler, mevsimsellik, döngüler ve rastgele dalgalanmaları analiz etmek için kullanılır (Ye et al. 2013). Bu analiz, geçmiş verilerden geleceğe yönelik tahminler yapmak ya da olaylar arasındaki ilişkileri incelemek için önemlidir. Zaman serisi analizi finans, ekonomi, radyasyon (Pala 2021), doğal gaz, güneş lekeleri (Pala and Atıcı 2019), iklim bilimleri (Pala and Şevgin 2024), sağlık (Pala, Atıcı, and Yıldız 2023) ve enerji (Pala 2023a, 2023b, 2024; Ünlük and Pala 2019) gibi birçok alanda uygulanır. Özellikle borsa fiyatları, satış tahminleri, hava durumu öngörülleri ve enerji tüketim trendleri gibi konularda karar destek sistemlerinde yaygın olarak kullanılır.

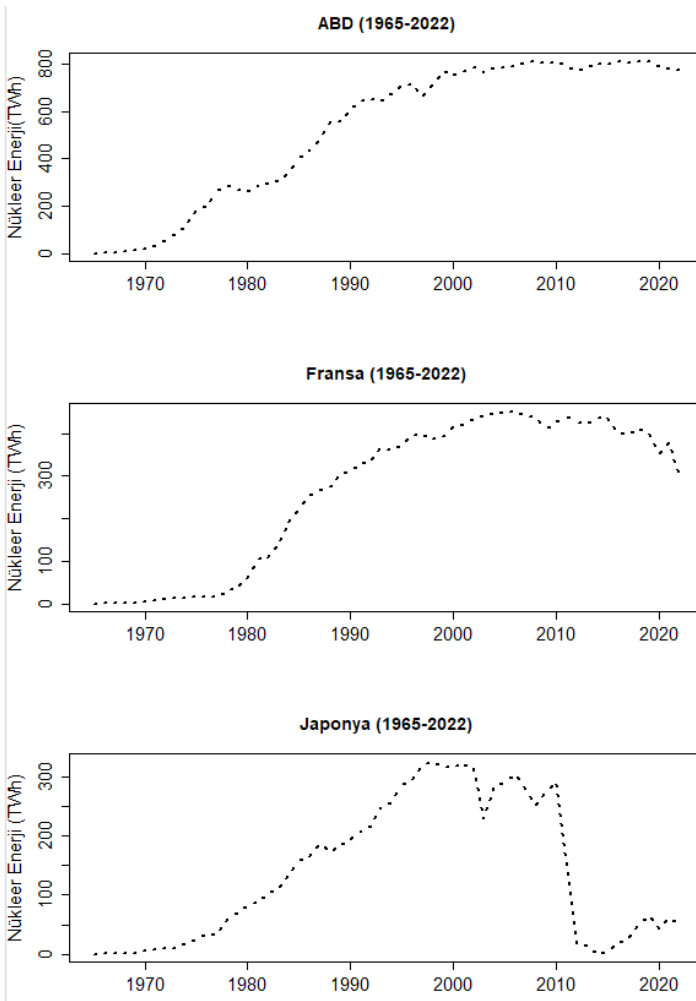
Amerika'nın nükleer enerjiden elektrik enerjisi üretiminde 1965-2022 yılları arasındaki eğilimleri anlamak için bu dönemdeki önemli artış ve azalışları dikkate almak gerekir. Amerika Birleşik Devletleri, küresel nükleer enerji üretimine önemli bir katkıda bulunmaktadır; toplam nükleer enerjinin üçte ikisinden fazlası Amerika Birleşik Devletleri, Fransa, Rusya, Güney Kore ve Çin tarafından üretilmektedir. Amerika Birleşik Devletleri ve Avrupa'da nükleer enerjiye yönelik olumsuz algıya rağmen, Amerika Birleşik Devletleri gibi hızlı büyüyen ekonomiler, sürekli elektrik arzı sağlamak için nükleer enerjiye güvenmeye devam ettiler. Amerika Birleşik Devletleri'nde nükleer üretimin ve sanayiye yönelik kamu desteğinin istikrarlı kalması dikkat çekicidir. Ek olarak, ABD ve Rusya'nın her ikisi de nükleer enerji ve nükleer silah üretimi sırasında ortaya çıkan büyük miktarda nükleer atığa sahiptir.

Şekil 1'de görüldüğü gibi (üst) ABD'nin 1965-1970 'lerde nükleer enerjiden elektrik enerjisi üretiminde belirgin bir artış gözlenmektedir. 1980-2010 dönemi boyunca nükleer enerji üretimi sürekli olarak artmış ve 2010 civarında zirveye ulaşmıştır. 2010 ve sonrası dönemlerde üretim seviyeleri zirveye ulaştıktan sonra sabit kalmış gibi görünmektedir.

Fransa, elektrik üretimine odaklanarak nükleer enerji üretiminde önemli bir oyuncu olmuştur. 1965 ile 2022 yılları arasında Fransa'nın nükleer enerji sektörü zorluklarla ve başarılarla karşı karşıya kaldı. Fransa'nın nükleer enerjiye olan bağlılığı, nükleer enerji kullanımındaki önemli artışla birlikte açıkça ortaya çıkmıştır. Ancak 2011 yılında meydana gelen Fukushima Daiichi Nükleer Santrali kazası gibi dış faktörler, nükleer enerjiye ilişkin

küresel algıyı etkileyerek nükleer enerjinin elektrik üretimindeki payının azalmasına yol açmıştır. Buna rağmen Fransa, AB'de ve dünyada lider nükleer güç konumunu korumayı hedefliyor. Ülkenin enerji geçiş politikası, 2035 yılına kadar %50 ila %40 oranında bir hedef ile nükleer ve yenilenebilir enerji kaynaklarını dengeleme planlarını içermektedir.

Şekil 1'de görüldüğü gibi (orta) Fransa'nın nükleer enerjiden elektrik enerjisi üretiminde 1970'lerin ortalarından itibaren hızlı bir artış gözlenmektedir. 1990-2010 aralığında üretim en yüksek seviyelerine ulaşmış ve sabit kalmıştır. 2010 ve sonrasında üretim seviyeleri kademeli olarak azalmış gibi görünmektedir.



Şekil 1: Üç farklı ülke için farklı zaman aralıklarında nükleer enerjiden elde edilen elektrik enerjisine ait zaman serisi grafikleri

Japonya'nın 1965'ten 2022'ye kadar nükleer enerji yörüngesi dalgalanmalara tanık oldu. Fukushima öncesi Japonya dünyanın üçüncü büyük nükleer kapasitesine sahipti, ancak 2011'deki felaket nükleer enerji üretiminde keskin bir düşüşe yol açarak 2014'te sıfır terawatt saate ulaştı. 2018 itibarıyla Japonya'nın enerji karışımında nükleerin payı yalnızca %4'tü. Küresel ölçekte nükleer elektrik üretiminin payı 1996'daki %17'den 2019'da yaklaşık %10'a düştü; birincil enerjinin payı ise %4 oldu. Japonya'nın nükleer payı 1998'deki %31'den 2014'ten sonra ihmal edilebilir bir düzeye düşmesine rağmen, ülke 2015'ten sonra reaktörleri yeniden başlattı. Fukushima sonrası nükleer enerjinin öneminin azalması, Almanya'da nükleer enerjinin 2022 yılına kadar aşamalı olarak ortadan kaldırılmasını hızlandırdı. Japonya'nın savaş sonrası enerji stratejisiyle iç içe geçmiş nükleer enerji politikası, önemli olayların ve politika değişimlerinin damgasını vurduğu karmaşık bir yolculuğu yansıtıyor.

Şekil 1'de görüldüğü gibi (alt) Japonya'nın nükleer enerjiden elektrik enerjisi üretiminde 1970'lerin sonlarından itibaren nükleer enerji üretiminde hızlı bir artış gözlenmektedir. 1990-2010 aralığında üretim en yüksek seviyelerine ulaşmıştır. 2011'deki Fukushima felaketinden sonra belirgin bir düşüş görülmektedir. Üretim seviyeleri önemli ölçüde azalmış ve düşük seviyelerde kalmıştır.

Burada yapılan zaman serileri kestirimleri için hem derin öğrenme hem de istatistiksel-tabanlı modeller kullanılmıştır. R programlama dili ortamında, R-tabanlı Naive, Ses, Auto.Arima, Holt-Winters, Ets, Thetaf, Nnetar ve Mlp tahmin modelleri kullanılmıştır.

Naive: Naive model, istatistiksel bazlı zaman serilerinin en basit tahmin modelidir. Bu model, gelecekteki değerleri tahmin etmek için yalnızca mevcut zaman serisi verilerini kullanır ve herhangi bir trend, mevsimsellik veya desen varsayımı yapmaz. Naive model, gelecekteki değerleri, genellikle geçmiş değerlerin son gözlem değeri veya bir ortalama ile tahmin eder.

Auto.Arima: Auto.Arima (AutoRegressive Integrated Moving Average - Otomatik Regresyonlu Entegre Hareketli Ortalama), R programlama dili içinde bulunan forecast paketinde yer alan bir fonksiyondur. Bu fonksiyon, otomatik olarak zaman serisi verilerine uygun ARIMA modellerini belirlemek için kullanılır. ARIMA modelleri, zaman serisi verilerindeki trendleri, mevsimsellikleri ve değişkenlikleri tanımlamak için

kullanılan istatistiksel modellerdir. Auto.Arima, verilen zaman serisi verilerini alır ve en uygun ARIMA modelini otomatik olarak belirler.

ETS: ETS (Exponential Smoothing – Üstel Düzeltme) modeli, zaman serilerinin tahmin edilmesi için kullanılan bir sınıf istatistiksel modeldir. Bu model Robert G. Brown tarafından 1950 yılında basit üstel düzleştirme yöntemini kesikli verilerle genişletmiş trend ve mevsimsellik için kullanışlı hale getirmiştir.

Holt-Winters: Holt-Winters yöntemi, zaman serisi verilerini modellemek ve gelecekteki değerleri tahmin etmek için kullanılan bir yöntemdir. Bu yöntem, bir zaman serisindeki düzenli trendler, mevsimsellik ve dönemsel değişkenler gibi desenleri yakalamak için tasarlanmıştır. Holt-Winters yöntemi, üç temel bileşeni içerir: düzey (level), trend ve mevsimselliktir. Bu bileşenler, zaman serisinin gelecekteki değerlerini tahmin etmek için kullanılır.

Ses: SES (Simple Exponential Smoothing - Basit Üstel Düzeltme) modeli, zaman serilerini tahmin etmek için yaygın olarak kullanılan bir yöntemdir. Bu yöntem, geçmiş verileri kullanarak gelecekteki değerleri tahmin etmeyi amaçlar.

THETAF: Theta modeli zaman serilerinde kullanılır. Model zaman serisindeki eğrilerin üzerine, teta kat sayısı adında yapılan değişikliklerin uygulanmasıyla elde edilir. Teta değeri sıfır ($Q=0$) olduğunda eğri doğrusal bir regresyon çizgisine dönüşür. Teta kat sayısının küçük olması deflasyon derecesini artırır, dalgalanmaların azaldığı ortamda verilerdeki uzun vadeli eğilimlerin daha belirgin olması sağlanır.

NNAR: NNAR (Neural Network Autoregressive - Otoresif Sinir Ağı) karmaşık veri yapılarını modellemek için kullanılan bir yapay zekâ modelidir. Genellikle ekonomi, finans, hava durumu tahmini ve diğer zaman serisi analizlerinde kullanılır.

MLP: MLP (Multilayer Perceptrons - Çok Katmanlı Algılayıcılar), yapay sinir ağlarından çok katmanlı ileri beslemeli bir modeldir. Mlp üç temel katmandan oluşur bunlar giriş katmanı, gizli katman ve çıkış katmanından oluşur. Her katman, önceki katmandan gelen girdileri alır, ağırlıklarla çarpar, bir aktivasyon fonksiyonuna sokar ve çıktıyı üretir. Eğitim sırasında, model, girdilerle gerçek çıktı arasındaki hatayı en aza indirmek için ağırlıkları ve sapmaları (bias) ayarlamak için geri yayılım algoritmasını kullanır.

İstatistiksel modeller, genellikle istikrarlı zaman serisi verileriyle iyi çalışırken, derin öğrenme modelleri daha karmaşık ve değişken veri yapılarına uyum sağlayabilir. İklimsel değişkenleri etkileyen birçok faktörün karmaşıklığı göz önüne alındığında, hem istatistiksel hem de derin öğrenme modellerinin birlikte kullanılması, veri heterojenitesi sorununu ele almak ve daha doğru tahminler sağlamaktır. Analizlerin tamamı Intel i5 PC 3.2 GHz CPU, 8 GB RAM, SATA disk ve Windows 10 Pro işletim sistemi konfigürasyonuna sahip bir bilgisayar yardımıyla gerçekleştirilmiştir.

ABD'nin 1965-2022 yılları arasındaki nükleer enerjiden elektrik üretim sürecinin gelecek tahminleri için ikisi derin öğrenme modeli ve altısı istatistiksel model olmak üzere toplamda sekiz model kullanılmıştır. Her bir model için veri setinin eğitim ve test işlemleri için bölünmesine bağlı olarak üçer tane analiz yapılmıştır. Analiz sonuçları hem eğitim hem de test için RMSE, MAE ve MAPE metrikleri yardımıyla değerlendirilmiştir.

RMSE (Hata Karelerinin Ortalamasının Karekökü): RMSE, modelin tahminlerinin gerçek değerlerden ne kadar uzak olduğunu ölçen bir ölçümdür. Genellikle regresyon modellerinin doğruluğunu değerlendirmek için kullanılır. Hata karelerinin ortalaması alındığı için büyük hataları daha fazla cezalandırır. Bu, modeli büyük hatalara karşı hassas hale getirir (Du et al. 2019)..

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

MAE: Tahmin hatalarının ortalama mutlak değerini ölçer. Regresyon analizlerinde modelin ortalama performansını değerlendirmek için kullanılır. MAE daha basit ve daha anlaşılırdır çünkü her hata mutlak değerler olarak ele alınır. Hata mutlak değerleri kullanıldığı için büyük hatalar üzerinde daha az etkisi vardır (Bisht and Ram 2021).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

MAPE: Tahminlerin orantılı olarak ne kadar yanlış olduğunu ölçer ve sıklıkla zaman serisi tahmini gibi uygulamalarda kullanılır. Hataların orantılı büyüklüğünü gösterir ve bu da farklı ölçeklerdeki verileri karşılaştırmayı kolaylaştırır (Singh and Yassine 2018).

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i|} \times 100 (\%) \quad (3)$$

ABD zaman serisi kullanılarak yapılan analizler Tablo 1’de verilmiştir. Tablo 1’de görüldüğü gibi ABD’nin 1965-2022 yılları arasındaki 58-yıllık nükleer enerjiden elektrik enerjisi üretimi için uzun vadeli olarak ifade edebileceğimiz tahminler yapılmıştır. Her bir model için eğitim/test uzunlukları sırasıyla 33/25, 31/27 ve 29/29 olarak ayarlanmıştır. Bu da sırasıyla %57/%43, %53/%47 ve %50/%50 olarak yüzdelik biçiminde ifade edilebilmektedir. Tabloda görüldüğü gibi her model için yapılan üçer analizdeki en iyi başarımlar kalın olarak gösterilmiştir. Veri setinin %57 ve %43 olarak ayrıldığı eğitim ve test oranlarında SES (Luo et al. 2017), Auto.Arima (Zhang 2003), Holt-Winter (Koehler, Snyder, and Ord 2001), ETS , Thetaf ve MLP (Widiasari, Nugroho, and Widyawan 2018) modelleri daha iyi tahminlerle ön plana çıkmışlardır. Veri setinin %53 ve %47 olarak ayrıldığı eğitim ve test oranlarında Thetaf modeli, %50 olarak ayrıldığı eğitim ve test oranlarında ise NNETAR modeli daha iyi tahminlerle ön plana çıkmıştır. Genel anlamda test verileri üzerinde yapılan en iyi tahmin NNETAR modeli tarafından verinin %57/%43 olarak ayrıldığı durumda %12,04 MAPE değeri ile yapılmıştır. MAPE değerinin düşük olması daha iyi tahmin anlamına gelmektedir. MAPE değerinin %10’dan küçük olması, yüksek doğruluklu tahmin; %10 ile %20 arasında olması İyi tahmin; %20 ile %50 arasında olması kabul edilebilir tahmin ve %50 den büyük olması zayıf yada yanlış tahmin olarak değerlendirilmektedir(Bisht ve Ram, 2021).

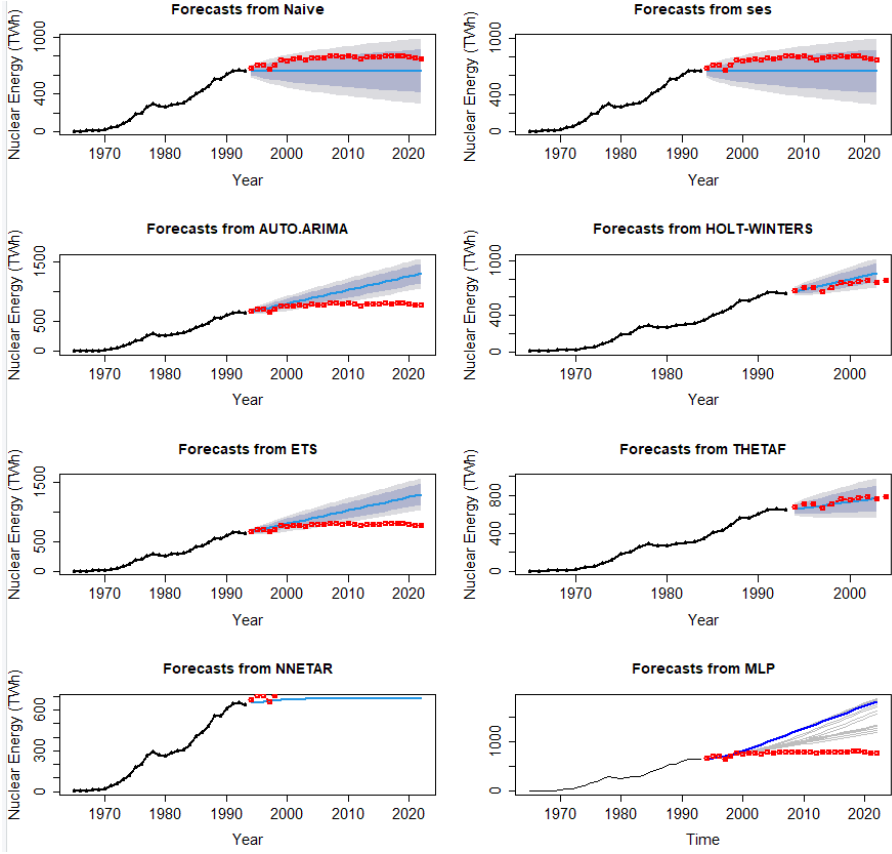
MAPE değeri yüzdelik olarak yapılan hatayı ifade etmektedir. Bu durumda THETAF modelinin başarımları %100-%15,15=%84,85 olarak hesaplanabilmektedir. MAPE metriğine uygun olarak elde edilen RMSE ve MAE değerleri de tahminin kabul edilebilirliğini desteklemektedir. Çünkü RMSE ve MAE değerlerinin küçük olması daha iyi tahmin anlamına gelmektedir. ABD ‘nin 1965-2022 yılları arasındaki nükleer enerjiden elektrik enerjisi üretim tahminlerinde kullanılan model grafikleri Şekil 2’de verilmiştir. Buradaki grafikler eğitim/test verilerinin %50 / %50 olarak alındığı durumlara ait tahmin grafikleridir.

Grafiklerdeki kırmızı çizgiler test verilerinin ortalamasını, mavi çizgiler test verisi uzunluğunda yapılan tahminlerin ortalamasını, koyu gri alanlar %80 olasılıkla tahmin aralığını ve açık gri bölgeler ise %95 olasılıkla tahmin

aralığını göstermektedir. Şekil 2’de gerçek ortalamaya en yakın ortalamalar görsel olarak THETAF ve HOLT-WINTERS modelleri tarafında yapılan tahmin grafiklerinde göze çarpmaktadır.

Tablo 1 ABD için 1965-2022 yılları arasında nükleer enerjiden üretilen elektrik enerjisine ait tahmin metrik değerleri

Veri seti	Eğitim/test setleri			
ABD (1965-2022)	Eğitim/test uzunluğu	RMSE	MAE	MAPE (%)
Naive	33/25	32,96/125,40	25,81/123,45	14,74/32,70
	31/27	32,87/76,99	25,86/72,48	15,47/19,20
	29/29	32,84/135,45	25,34/129,46	16,23/34,29
SES	33/25	32,46/125,39	25,03/123,44	14,34/15,65
	31/27	32,34/76,99	25,03/72,48	15,02/16,204
	29/29	32,27/135,45	24,47/129,46	14,92/16,53
Auto,Arima	33/25	24,28/174,00	18,69/136,52	24,81/27,29
	31/27	22,61/306,84	18,38/260,00	37,33/39,98
	29/29	23,22/268,65	18,89/214,66	38,60/37,21
Holt-Winters	33/25	25,77/158,54	19,63/151,89	18,02/19,54
	31/27	22,87/98,75	19,01/89,67	53,44/61,89
	29/29	23,55/52,69	19,58/42,69	56,69/65,80
ETS	33/25	25,77/158,54	19,63/151,89	18,02/28,48
	31/27	22,87/306,98	19,01/260,12	53,44/32,99
	29/29	23,55/263,31	19,57/210,02	56,65/26,62
Thetaf	33/25	32,46/44,45	25,03/39,48	14,34/15,15
	31/27	32,34/38,20	25,03/32,30	15,02/14,39
	29/29	32,27/28,04	24,47/24,78	15,70/16,40
Nnetar	33/25	23,43/96,43	17,86/95,00	10,06/12,04
	31/27	22,05/46,55	17,01/43,75	10,68/15,58
	29/29	22,43/96,17	16,67/91,32	10,96/15,98
Mlp	33/25	22,79/253,22	16,67/195,68	19,79/24,74
	31/27	19,45/670,44	14,43/577,92	32,93/73,34
	29/29	20,05/540,75	14,91/424,97	28,91/53,74

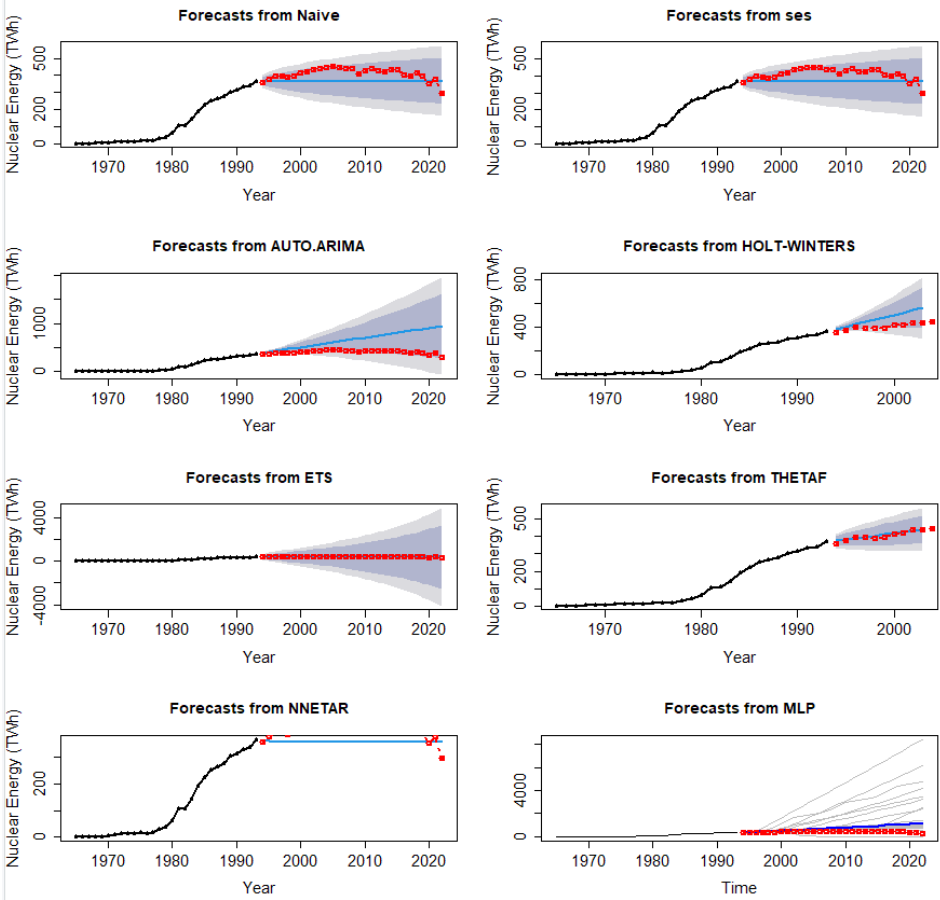


Şekil 2: ABD için verinin eğitim ve test için sırasıyla %57 ve %43 olarak alındığı durumda elde edilen model grafikleri

Tablo 2’de verildiği gibi Fransa’nın nükleer enerjiden elektrik enerjisi ürettiği veri setinin analizi neticesinde modellerin bir kısmı (Naive, SES, Holt-Winters ve NNTAR) 33/25 eğitim/test oranında başarılı olurken diğer bir kısmı 31/27 eğitim/test oranında (Holt-winter, Thetaf ve Nnetar) başarılı olmuştur. RMSE, MAE ve MAPE metrikleri birlikte değerlendirildiğinde en başarılı sonuç THETAF modeli tarafından gerçekleştirilmiştir. Bu durumu Şekil 3’de verilen grafiklerde de görmek mümkündür.

Tablo 2 Fransa 1965-2022 yılları arasında nükleer enerjiden üretilen elektrik enerjisine ait tahmin metrik değerleri

Veri seti	Eğitim/ Test setleri			
Fransa (1965-2022)	Eğitim/Test uzunluğu	RMSE	MAE	MAPE (%)
Naive	33/25	18,80/39,10	13,11/32,59	17,00/18,01
	31/27	19,06/49,08	13,25/43,95	17,95/19,56
	29/29	19,39/54,18	13,29/48,68	18,99/21,66
Ses	33/25	18,51/39,10	12,72/32,59	16,53/18,01
	31/27	18,75/49,08	12,83/43,95	17,41/20,56
	29/29	19,06/54,18	12,83/48,69	18,38/21,66
Auto.Arima	33/25	12,51/130,21	8,88/97,29	11,42/25,18
	31/27	12,47/175,32	8,65/135,71	11,96/34,66
	29/29	11,71/312,28	8,06/257,94	12,50/64,88
Holt-Winter	33/25	11,97/21,48	8,27/18,48	12,77/14,32
	31/27	11,97/10,28	8,14/8,83	12,90/12,93
	29/29	11,60/78,15	7,93/70,28	12,84/17,08
Ets	33/25	11,97/127,42	8,27/96,19	12,77/24,88
	31/27	18,75/49,08	12,83/43,95	17,68/20,56
	29/29	19,06/54,18	12,84/48,69	18,68/21,66
Thetaf	33/25	18,51/12,60	12,72/8,92	16,53/22,11
	31/27	18,75/7,58	12,83/6,05	12,41/12,48
	29/29	19,06/9,62	12,83/7,58	18,38/21,94
Nnetar	33/25	10,40/47,00	7,38/41,54	11,49/12,00
	31/27	10,17/58,28	7,02/53,57	12,14/12,78
	29/29	10,09/63,22	6,69/57,51	12,74/13,70
Mlp	33/25	5,41/60,11	3,45/40,87	31,25/40,91
	31/27	5,95/312,37	3,80/270,05	32,51/67,56
	29/29	3,35/393,09	2,35/311,73	28,66/78,68

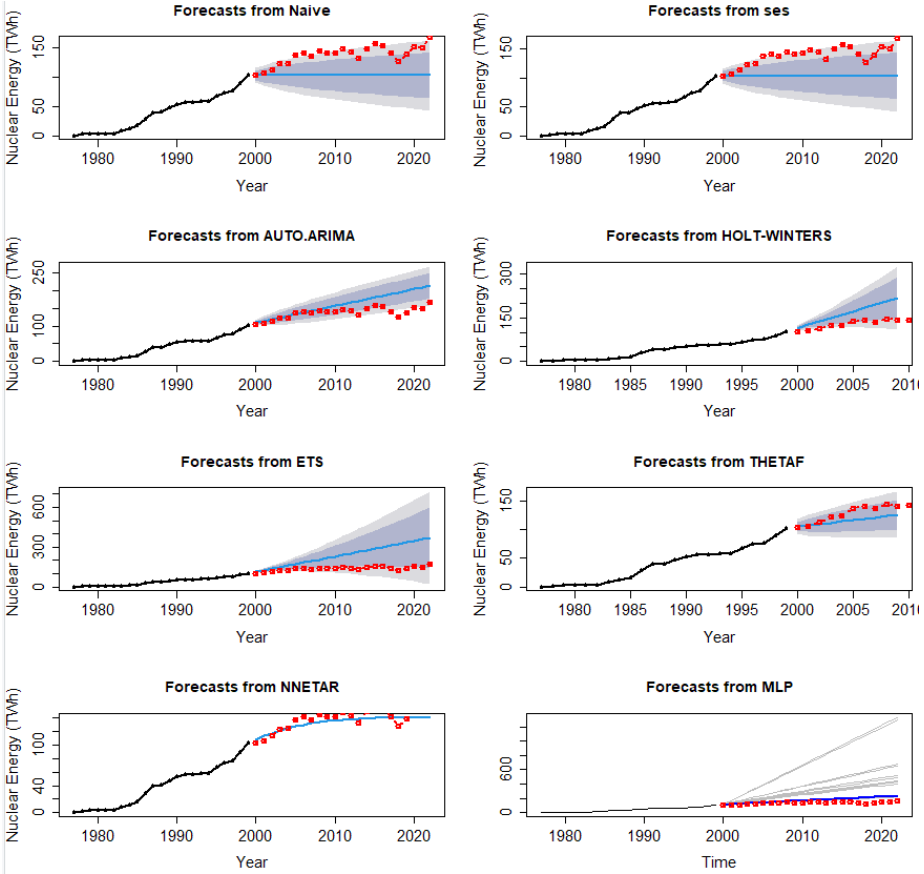


Şekil 3: Fransa için verinin eğitim ve test için sırasıyla 31 ve 27 uzunlukta alındığı durumda elde edilen model grafikleri

Japonya'nın elektrik enerjisi üretimi için yapılan analiz sonuçları Tablo 4.4'te verilmiştir. Japonya'nın 1965 ile 2022 yılları arasındaki nükleer enerji üretimine ilişkin zaman serisindeki ani düşüş ve yükselişler tahmin modellerinin performansını da olumsuz etkilemiştir. Özellikle 2011 yılındaki Fukushima felaketi üretimin tamamen durmasına neden olmuş bu da tahmin modellerinin başarımlarını olumsuz etkilemiştir. Modellerden daha iyi performans elde etmek için veri setinin logaritması alındıktan sonra tahmin analizleri yapılmıştır. Yapılan analizlerde Auto.Arima modelinin daha iyi tahmin ile ön plana çıktığı gözlenmiştir.

Tablo 3 Japonya'nın 1965-2022 yılları arasında nükleer enerjiden üretilen elektrik enerjisine ait tahmin metrik değerleri

Veri seti	Eğitim/test veri setleri			
Japonya (1965-2022)	Eğitim/test uzunluğu	RMSE	MAE	MAPE (%)
Naive	44/14	0,54/2,86	0,24/2,17	49,51/84,23
	43/15	0,55/2,84	0,24/2,10	50,64/81,23
	42/16	0,56/2,81	0,25/2,05	51,84/78,57
Ses	44/14	0,54/2,86	0,23/2,17	48,39/84,23
	43/15	0,54/2,84	0,24/2,10	49,47/81,23
	42/16	0,55/2,81	0,24/2,05	50,61/78,57
Auto.Arima	44/14	0,28/2,98	0,18/2,29	48,95/88,24
	43/15	0,25/3,28	0,18/2,62	46,94/96,75
	42/16	0,25/2,49	0,18/1,64	26,96/35,71
Holt-Winter	44/14	0,38/3,36	0,21/2,53	70,08/96,63
	43/15	0,38/3,40	0,22/2,44	72,15/85,44
	42/16	0,40/3,36	0,21/2,26	80,24/90,25
Ets	44/14	0,27/3,01	0,16/2,33	31,63/89,42
	43/15	0,30/2,99	0,15/2,30	39,80/87,08
	42/16	0,32/2,90	0,16/2,16	40,28/81,88
Thetaf	44/14	0,54/3,60	0,23/2,78	48,39/85,91
	43/15	0,54/3,67	0,24/2,73	49,47/88,96
	42/16	0,55/3,67	0,24/2,61	50,61/92,12
Nnetar	44/14	0,23/2,83	0,14/2,14	17,16/83,13
	43/15	0,23/2,75	0,14/2,01	17,51/78,19
	42/16	0,24/2,68	0,14/1,90	17,91/73,63
Mlp	44/14	0,11/2,92	0,08/2,25	6,29/86,59
	43/15	0,11/3,11	0,07/2,42	6,38/91,00
	42/16	0,11/2,98	0,08/2,24	6,32/84,43



Şekil 1 Japonya için verinin eğitim ve test için sırasıyla 42 ve 16 uzunlukta alındığı durumda elde edilen model grafikleri

SONUÇ

ABD verisinde veri seti eğitim/test için %57'ye %43 olarak ayarlandığında en başarılı model %87,96 MAPE doğruluk oranıyla NNETAR modeli olmuştur. Bunu sırasıyla, THETAF ve SES modelleri takip etmiştir. Fransa verisinde veri seti eğitim ve test oranları %53'e %47 oranında ayrılmıştır. Burada en başarılı model %87,52 Mape doğruluk oranıyla THETAF modeli olmuştur. Bunu sırasıyla NNETAR ve Holt-Winters takip etmiştir. Japonya verisinde 2011 yılında Japonya'daki nükleer patlama veri analizini olumsuz bir şekilde etkilediği için bu örnekte verilerin logaritması alınarak yeni bir veri seti oluşturulmuştur. Bu veri setinde ise veriler eğitim/test oranı %72'ye %28 oranında ayrılmıştır. En başarılı model %64,29 MAPE doğruluk oranıyla Auto.arima modeli olmuştur. Bunu sırasıyla NNETAR ve SES takip etmiştir.

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Fire – Smoke Classification using Machine Learning Methods

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ABSTRACT

Although fires are often perceived as regional events due to their impact on specific areas, their overall consequences extend far beyond those regions. For instance, forest fires not only devastate forests, pastures, and crops, but also result in the death of wildlife and livestock, cause damage to human settlements, and endanger the lives of residents in the affected areas. Consequently, their impact reaches significantly beyond the immediate region. The restoration of these forests is a complex process that poses substantial challenges to the national economy. Therefore, early fire detection and the implementation of preventive measures, or swift intervention, are of critical importance. The classification of fire images is a key area of study supporting all these preventive measures. In this study, fire image classification was performed using machine learning techniques, specifically Random Forest and Logistic Regression models. To improve the feature extraction process, image embeddings derived from the Inception V3 model were used. The analysis was conducted using Python-based Orange 3 software. Based on the results obtained from analyzing 12,000 data points across three classes (fire, non-fire, smoke), machine learning methods have proven to be highly effective in image processing.

Keywords – Fire detection; machine learning; image classification; random forest; logistic regression; image embedding

I. INTRODUCTION

According to the sixth report of the Intergovernmental Panel on Climate Change (IPCC), global warming has reached critical levels [1]. The report reveals that a temperature increase of 1.1°C above pre-industrial levels has been observed over the last 200 years. This rise is already having a detrimental impact on human life. If the warming trend continues, the 1.5°C threshold will likely be surpassed, causing severe harm to all living organisms and resulting in more frequent and dangerous weather events [2]. Fires, as one of the major disasters, have detrimental effects on communities, individuals, and the environment. Reducing risk, implementing preventive measures, and preparing for potential fires are of paramount importance. The causes of forest fires and urban fires differ significantly. Forest fires can be triggered by human activities, such as arson and campfires, or by natural events like lightning strikes. In contrast, most urban fires result from human actions, such as cooking accidents or faulty electrical systems [3].

Fire classification is crucial for the immediate control and detection of fires. It plays an essential role in early detection and the planning of effective fire management and response strategies. Early fire notifications enable

individuals to evacuate the area and seek emergency assistance, helping to prevent extensive damage to property or assets. Such notifications also allow for prompt action to mitigate the fire's impact. While traditional fire alarm systems rely on smoke or heat sensors, fire detection using computer vision can cover larger areas and provide earlier warnings [4].

Airborne and spaceborne remote sensing methods are vital for establishing global early warning systems to forecast and prevent forest fires or manage them with minimal consequences [5]. Consistently collected data from these methods, in raster or vector form, can assist in detecting and reporting potential causes of forest fires before they escalate [6].

Computer-aided data classification studies offer decision support to experts across various fields, ranging from text mining [7], [8], [9], [10] to vision-based studies. Numerous computer-assisted investigations have been conducted to identify forest fires. Advancements in artificial intelligence have facilitated the development of computer-assisted models that support research on fire classification and other related areas [9].

Machine learning algorithms have revolutionized fire classification and have gained increasing significance in research. Fires are often considered uncontrollable and unpredictable in regions with combustible vegetation, posing substantial threats to both the environment and the economy. The rising risk of severe fires, exacerbated by climate change and human activities, highlights the critical need for early detection methods. In this context, machine learning and deep learning techniques provide effective systems for fire identification [11].

Recent fire classification studies have shown that deep learning-focused approaches are used more frequently than traditional methods. This study investigates whether traditional classification techniques can produce accurate results that rival modern methods. Traditional data classification methods remain valuable, as they require less data, operate in shorter time frames, and place less strain on computer processors[12]. In this study, fire-image classification analyses are conducted using traditional machine learning techniques.

The structure of this paper is as follows: Section II provides a brief overview of existing research on fire classification using machine learning. Section III describes the materials and methods employed in this study. Section IV presents the experiments conducted and their corresponding results. Finally, Section V concludes with a summary of the findings and final remarks.

II. LITERATURE REVIEW

The classification of forest fires or fires in urban areas has been a subject of study from past to present [13]. It is based on categorizing the image according to predetermined criteria and measures. Vector clusters and pixel information of the image are utilized, and each of the spectral and/or textural features, or

their various combinations, can be employed in the classification. In this way, the image is classified into the relevant category [14]. The classification of forest fires enables managers to both analyze the cause of the fire and develop procedures for actions before, during, and after the fire. For instance, while images from an automatic system provide information about the fire, if it is determined that the fire was caused by land conversion or degradation, the appropriate procedure can be applied.

Arnett et al. estimated and classified fire severity to map fire effects and assess damage to the woody surface of forests [15]. By determining the most suitable features for the model, Harkat et al. demonstrated that machine learning techniques can compete with the deep learning methods that have recently gained popularity [16]. In the analysis, two publicly available datasets were utilized: the Corsican dataset with 1,175 fire images and the FLAME dataset with 2,003 fire images and 2,842 non-fire images. Additionally, one private dataset, the Firefront_Gestosa dataset, containing 238 fire images and 1,113 non-fire images, was used to classify fire and non-fire pixels using the Support Vector Machine (SVM) approach. An overall accuracy of 96.21% was achieved.

Chiano et al. proposed a new method called BoWFire to detect fire in still images. In their study using 226 images, Naïve Bayes and KNN classifiers were employed to classify fire images, with fewer parameters and false positives discarded [17]. Dampage et al. classified fire images consisting of 7,000 data points as either fire or non-fire. They selected features such as light intensity, temperature, and CO levels from a wireless sensor network and proposed a machine learning regression model along with threshold ratio analysis [18]. Koetz et al. classified land cover for forest fire management by combining two remote sensing systems: light detection and ranging (LiDAR) and imaging spectrometry. The SVM method was chosen for its high-dimensional, multi-source, and non-parametric capabilities. The study achieved an accuracy of 75.4% with 1,248 independent samples [19].

In the study proposed by Rosadi et al., data consisting of 202 topographic and meteorological variables from the Kalimantan Selatan Province, classified as fire or non-fire, were used. Variables such as time, district area, land surface temperature, wind speed, humidity, altitude, and the normalized vegetation index were included. They proposed Support Vector Machines (SVM), k-Nearest Neighbor (kNN), Logistic Regression (LR), AdaBoost Decision Tree (DT), and Naïve Bayes (NB) as classification tools. The AdaBoost method demonstrated superiority over the other methods [20].

In this study, 12k images labeled as fire, smoke, or non-fire were used from datasets available on Kaggle [21]. Logistic Regression and Random Forest methods were used to classify the images as fire/smoke or fire/non-fire.

III. MATERIAL AND METHOD

A. Data Gathering

The Fire and Smoke dataset used in this study consists of a total of 12k fire, smoke, and non-fire images uploaded to Kaggle by Abdullah Butt, currently divided into training and test sets [21]. Fig. 1 shows sample images from the dataset. The rationale for using this dataset includes the fact that academic studies have yet to utilize it, as previous research primarily focused on binary classification, distinguishing between fire and non-fire images. In contrast, this dataset provides images for three distinct classes: fire, smoke, and non-fire. In summary, using image processing without relying on additional weather or geographical information enables researchers to develop fast and efficient fire/smoke detection systems.

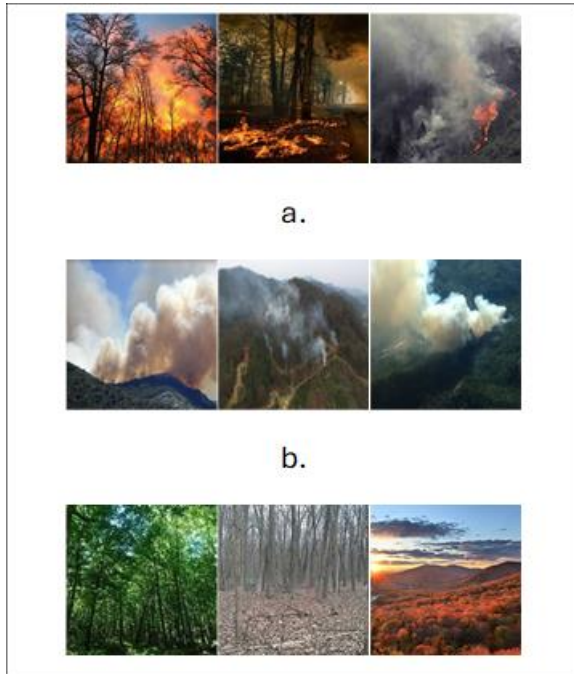


Fig. 1. Sample images of dataset (a. fire, b. smoke, and c. non-fire)

B. Classification Methods

Logistic Regression (LR) and Random Forest (RF), two widely used techniques with a proven track record in classification tasks, are employed in this study. The choice of these techniques is based on their extensive use in the literature. A summary of these methods is provided below.

a) Logistic Regression

Logistic regression is one of the most widely used methods in statistical modelling and data analysis. It determines the relationship between the dependent variable and the independent variables in a model with categorical (especially binary) dependent variables (e.g., yes/no or true/false) [22]. Although similar to the linear regression model, it differs in that the dependent variables do not have a continuous structure. It is employed when linear regression is insufficient. The probability of the independent variables belonging to a specific category is estimated using the logit function. The significance of the model is assessed by statistically calculating the effects of the independent variables on the dependent variable. For example, if the coefficient of an independent variable is positive, an increase in this variable raises the likelihood of the dependent variable being 1. Logistic regression is typically applied using the Maximum Likelihood Estimation (MLE) method. This method estimates the model parameters in a manner that maximizes the likelihood of the observed data [23].

b) Random Forests

Random Forest is a powerful classification and regression method widely used in machine learning. This model combines multiple decision trees and is one of the ensemble learning methods designed to produce more accurate and stable results by aggregating the predictions of each tree. The ensemble learning approach is based on combining multiple models to create a more robust model. Decision trees are generated using a random subset of the training data, a technique known as bootstrap sampling. Additionally, a random subset of features is chosen at each node to determine the best split. These two sources of randomness help prevent overfitting and improve the model's generalization ability. In classification problems, the predictions of each tree are aggregated, and the class with the most votes is selected [24], [25].

c) Model Construction and Environment

The related methods were applied to the dataset using the Python-based Orange 3 program, an open-source machine learning and data visualization package. The main reasons for choosing Orange 3 include its ability to easily upload data without the need for coding, its data analysis capabilities, and the fact that it is free software with a variety of machine learning and data visualization features. Fig. 2 shows the model configuration used for training the proposed classification methods. In this configuration, the first step involves importing images for the training process. In this step, 3k fire, 3k smoke, and 3k non-fire labeled images are imported. Next, the image embedding process is applied.

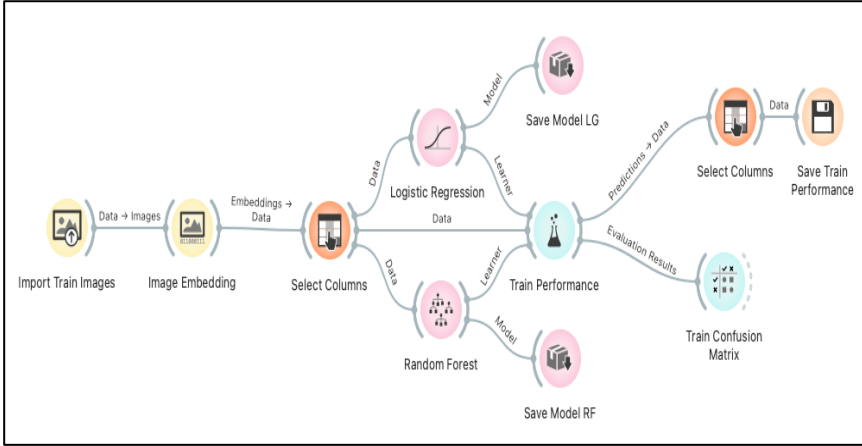


Fig. 2. Training configuration of proposed models using Orange 3

Orange 3 offers various image embedding methods, such as Inception V3, SqueezeNet, and VGG-16. In this study, Inception V3 is used as the default. Inception V3 is a convolutional neural network architecture from the Inception family that converts an image into a vector of numerical data. Each image in this study is transformed into 2,048 numerical features through image embedding, thus adding 2,048-dimensional features to each image. Afterward, input and target columns are selected from the available data using the Select Columns method. This process sets 2,048 numeric inputs (from image embedding) and three categorical outputs (fire, smoke, and non-fire). Logistic Regression and Random Forest methods are then trained separately using the same data, and the trained models are saved. Finally, the performance of the trained models is evaluated, and confusion matrices along with performance results are saved for interpretation. After completing the training process, the Orange 3 configuration shown in Fig. 3 is applied to evaluate the models' performance on the test dataset in a similar manner.

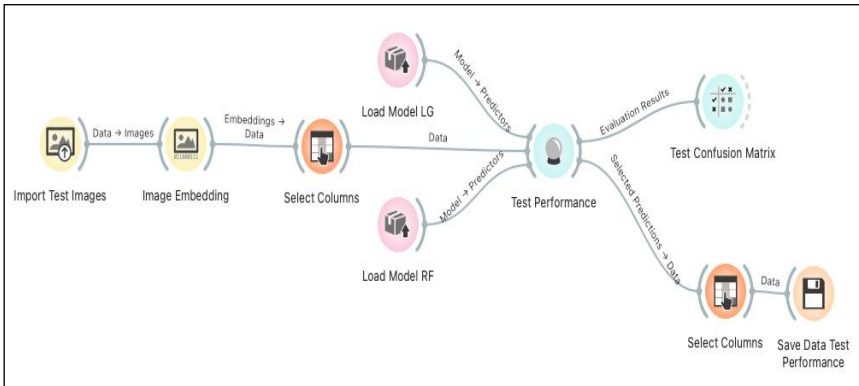


Fig. 3. Test configuration of proposed models using Orange 3

IV. EXPERIMENTAL STUDY AND RESULTS

The performances of the LR and RF methods trained on the Fire_and_Smoke dataset are evaluated. Various metrics are used to assess the performance of algorithms in classification tasks. In particular, the accuracy of identifying any given class is an important consideration. For this, performance indicators such as True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) should be calculated first. Based on these values, the True Positive Rate and False Positive Rate are calculated as shown in equations 1 and 2.

$$\text{True Positive Rate (TPR)} = TP/(TP+FN) \quad \text{Eq. (1)}$$

$$\text{False Positive Rate (FPR)} = FP/(FP+TN). \quad \text{Eq. (2)}$$

AUC (Area Under the Curve) is another metric used to evaluate the performance of a classification model. This metric calculates the area under the ROC (Receiver Operating Characteristic) curve to assess the model's accuracy, as shown in equation 3. If the AUC value is greater than 0.5, the model performs better than random predictions. The closer the AUC value is to 1, the better the model's classification performance.

$$AUC = \int_0^1 TPR(FPR)d(FPR) \quad \text{Eq. (3)}$$

Classification Accuracy (CA) is a key metric for measuring the accuracy of a machine learning model. It is calculated as the ratio of correctly predicted instances to the total number of instances. CA is evaluated using equation 4.

$$CA = (TP+TN)/(TP+FP+TN+FN) \quad \text{Eq. (4)}$$

CA alone may not be sufficient, and it is recommended to evaluate model performance in combination with other metrics such as F1-Score, Precision, Recall, and AUC.

The F1-Score is a metric used to evaluate a model's performance by balancing Precision and Recall. It is particularly important for assessing the trade-off between Precision and Recall, especially in unbalanced datasets. The F1-Score is calculated using equation 5.

$$F1 = 2*(Precision*Recall)/(Precision+Recall). \quad \text{Eq. (5)}$$

Recall (Sensitivity): Indicates how many true positive samples are correctly predicted and is calculated as shown in equation 6.

$$Recall = TP/(TP+FN)$$

Eq. (6)

MCC (Matthews Correlation Coefficient) is a metric used to evaluate the performance of a classification model. It is known for its ability to fairly measure model performance, even in the presence of class imbalance in classification problems. The formulation of MCC is provided in equation 7. The value of MCC can be interpreted as follows:

MCC = +1: The model makes a perfect classification (all samples are correctly classified).

MCC = 0: The model makes random predictions (predictions are purely by chance).

MCC = -1: All predictions made by the model are incorrect (all classifications are wrong).

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Eq. (7)

Table I presents the performance indicators for the proposed models on the training set. All performance indicators for both models show highly positive values, indicating that the training phase of the proposed classification methods has been quite successful.

Table 1. Train performance of the proposed models

Model	AUC	CA	F1	Precision	Recall	MCC
Random Forests	0.999	0.998	0.998	0.998	0.998	0.998
Logistic Regression	0.999	0.998	0.998	0.998	0.998	0.998

A Confusion Matrix is a table used to evaluate the performance of a classification model, providing detailed classification results. It helps to understand the types of errors made and the number of correct classifications by comparing the model's predicted results with the actual outcomes. Fig. 4 and Fig. 5 present the confusion matrices of the proposed models on the training and test sets, respectively.

		Predicted			
		Smoke	fire	non fire	Σ
Actual	Smoke	2995	5	0	3000
	fire	5	2995	0	3000
	non fire	0	0	3000	3000
Σ		3000	3000	3000	9000

		Predicted			
		Smoke	fire	non fire	Σ
Actual	Smoke	2995	5	0	3000
	fire	5	2995	0	3000
	non fire	0	0	3000	3000
Σ		3000	3000	3000	9000

a. CM for LR on training data set

b. CM for RF on training data set

Fig. 4. Confusion matrix of proposed algorithms on training set

		Predicted			Σ
		Smoke	fire	non fire	
Actual	Smoke	956	7	37	1000
	fire	18	979	3	1000
	non fire	26	25	949	1000
Σ		1000	1011	989	3000

		Predicted			Σ
		Smoke	fire	non fire	
Actual	Smoke	986	2	12	1000
	fire	19	969	12	1000
	non fire	39	27	934	1000
Σ		1044	998	958	3000

a. CM for LR on test data set

b. CM for RF on test data set

Fig. 5. Confusion matrix of proposed algorithms on test set

Both models misclassified only five smoke images as fire and five fire images as smoke in the training set. Given that smoke is present in many fire images and smoke images may also contain fire, this result is quite acceptable. Machine learning algorithms typically exhibit very high performance on training sets. However, it is crucial that algorithms also perform well on data not included in the training set. Therefore, the trained algorithms are evaluated on test sets to observe their performance. Table 2 presents the performance of the proposed algorithms on the test set, where all performance indicators are observed to be quite good.

Furthermore, Fig. 5 presents the confusion matrix for the test set. It can be observed that both models made the most errors when classifying non-fire images. Additionally, LR misclassified more smoke images, while RF misclassified more fire images compared to the other model. Despite these misclassifications, both models achieved CA values of 96%, demonstrating their ability to accurately identify fire, smoke, and non-fire images with a very acceptable level of consistency.

V. DISCUSSIONS AND CONCLUSIONS

In this study, Logistic Regression (LR) and Random Forest (RF), two widely used and easily applicable methods in the literature, were employed to classify the dataset presented by Butt on Kaggle, which is divided into 9k training and 3k test sets. The proposed machine learning algorithms were modeled using Python-based Orange 3 on the dataset, balanced across all three classes in both the training and test sets, and the results were analyzed.

A review of the training set performance suggests that the models demonstrate near-perfect accuracy. However, when examining the models' test performance, although the results remain high, they are relatively lower compared to the training set, as indicated by the confusion matrix and other performance metrics. Furthermore, although neither model shows significant superiority in the training set, the RF method slightly outperforms the LR method on the test set.

In conclusion, this study, which utilized traditional machine learning methods to classify fire, smoke, and non-fire images, demonstrates that the results can be readily applied and further refined to enhance technologies for detecting wildfire incidents—whether naturally occurring or human-induced—that are increasing globally due to climate change.

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Assessment of Rainfall Data Imputation Techniques: A Comparative Study with Focus on Thai Rainfall Dataset

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ABSTRACT

Addressing missing rainfall data remains a persistent complexity in hydrological modeling, water resource planning, and management. The need for comprehensive and reliable data is critical to allowing efficient water resource planning and management efforts. To address this issue, this study estimates missing values within three distinct datasets characterized by varying degrees of data incompleteness: Set-1 (10% missing data), Set-2 (30% missing data), and Set-3 (50% missing data). Employing various estimation methods, the investigation aimed to identify optimal approaches for accurate imputations. Using data from 31 observation stations in northern Thailand from 1993 to 2022, methods such as arithmetic averaging, long short-term memory recurrent neural network, multiple linear regression, M5 model tree, nonlinear iterative partial least squares, and K-nearest neighbor were evaluated. Among the rainfall stations, Tha Wang Pha and neighboring stations were selected based on Pearson's correlation. Three datasets with different levels of missing data were created. MLR consistently demonstrated strong performance, particularly in Set-1 with a high R^2 (0.94). In Set-2, MLR and LSTM-RNN showed the highest R^2 and lower error metrics, with MLR exhibiting an R^2 of 0.92 and LSTM-RNN an R^2 of 0.91. Despite 30% missing data in Set-2, MLR and LSTM-RNN remained robust. In the most challenging scenario, Set-3 with 50% missing data, LSTM-RNN excelled, exhibiting the highest R^2 of 0.90 and low error metrics, benefiting from its ability to capture temporal patterns. These results underline the importance of selecting appropriate models tailored to station characteristics for accurate prediction and monitoring in hydrological studies.

Keywords – Missing Data, Machine Learning, Rainfall, Artificial Intelligence, Imputation

I. INTRODUCTION

The absence of data within the rainfall dataset has engendered a spectrum of challenges. [1]. Neglecting to handle missing data may result in negotiated analyses, initiating the ability for biases and inaccuracies [2]. The accuracy of hydrological models significantly increased by adopting appropriate methods for analyzing missing rainfall data [3]. Hence, investigating and implementing appropriate methods for assessing missing values. It validates the attainment of thorough and dependable data, enabling robust analyses and precise conclusions. Several authors have utilized different estimation methods for imputing missing rainfall data [4-9]. Therefore, optimizing the estimation of missing data is optimal and feasible [5]. Several estimation methods have been developed for imputing missing data, which can be categorized into empirical, statistical, and function-fitting practices. Furthermore, methods involve estimating missing values based on observations commencing surrounding stations. Over the past two decades, estimation methods, such as decision trees (M5-MT) and neural networks (LSTM-RNN) have experienced notable recognition in the hydrological investigation. [10-12]. These methods accurately handle the uncertain and non-linear characteristics in hydrological statistics, demonstrating promising outcomes in the imputation of rainfall data. It is worth noting that many statistical analyses and AITs typically require complete datasets, unlike those containing missing values. Neglecting missing data is not advisable, as it results in the loss of valuable information and diminished inferential capabilities. [13]. Based on the dataset employed to construct the imputation models, estimation techniques can be broadly categorized into two groups. The first grouping comprises creating an approximation model that relies on spatial data. This evaluation model is typically utilized to predict missing rainfall data worldwide. It proves remarkably beneficial in scenarios where numerous stations have missing values. Moreover, the estimation of this data must be performed simultaneously for multiple stations, commonly referred to as targeted stations [14, 15]. In the second category, an estimation model is developed using historical data from the chosen target and neighboring stations. The model focused on the local scale; this model analyzes data from a limited number of rain gauge stations. This model is focused on the local scale, utilizing a limited number of rain gauge stations in the analysis. The estimation model is developed in this category by leveraging the historical daily rainfall data records from a target station and the closest nearby stations to develop a linkage between them. [4, 16, 17]. The current work in this study aligns with the second category. The subsequent discussion highlights some notable studies utilizing AITs for estimating missing rainfall data. Using machine learning, statistical techniques, and spatial interpolation, researchers. Thailand has tackled the problem of predicting missing daily or monthly rainfall amounts in various places [11, 18, 19]. Pinthong et al. (2022), conducted research in which the researchers explored

ML and spatial interpolation methods to estimate missing monthly data on rainfall. GP demonstrated an effective outcome among ML techniques. The authors underscored the importance of considering a correlation threshold of 0.80 or higher between the target and neighboring stations when employing these methods [11]. While the existing estimation methods have significantly contributed to advancing data imputation, there remains a need to explore alternative techniques to deliver the best performance in regions when minimal correlation exists between the target station and distant observation stations or rainfall stations, further research is necessary to devise more effective methods for accurately and reliably estimating missing rainfall data in these regions. They present unique challenges for data imputation.

Therefore, the main goal of the research is to perform a comparative analysis of several estimating techniques, including AITs like the long short-term memory recurrent neural network (LSTM-RNN) and the M5 model tree (M5-MT), alongside statistical methodologies including arithmetic averaging (AA), multiple linear regression analysis (MLR), and the non-linear iterative partial least squares (NIPALS) algorithm. This comparison will be specifically performed for the TMD target and neighboring stations. The subsequent sections delve into the study area, data, and imputation methods and provide a concise overview of the models. Results are presented in Section 3, while a detailed discussion and conclusion are presented in Section 4.

II. MATERIALS AND METHOD

1.1. Study Area

Thailand is situated between 5°37' and 20°27' north latitude and 97°22' and 105°37' east longitude [20]. Thailand is partitioned into five distinct climatic zones, each characterized by unique hydrological traits influencing the region's water resources and hydrological dynamics [21]. In this study, data from 31 TMD rainfall stations situated in the northern region were employed. The northern region experiences a tropical monsoon climate with three discernible seasons: a cool season spanning November to February, a rainy season from June to October, and a humid season from March to May. The rugged terrain and steep slopes in this area accelerate erosion and drainage, particularly during the rainy season. [20, 22-25].

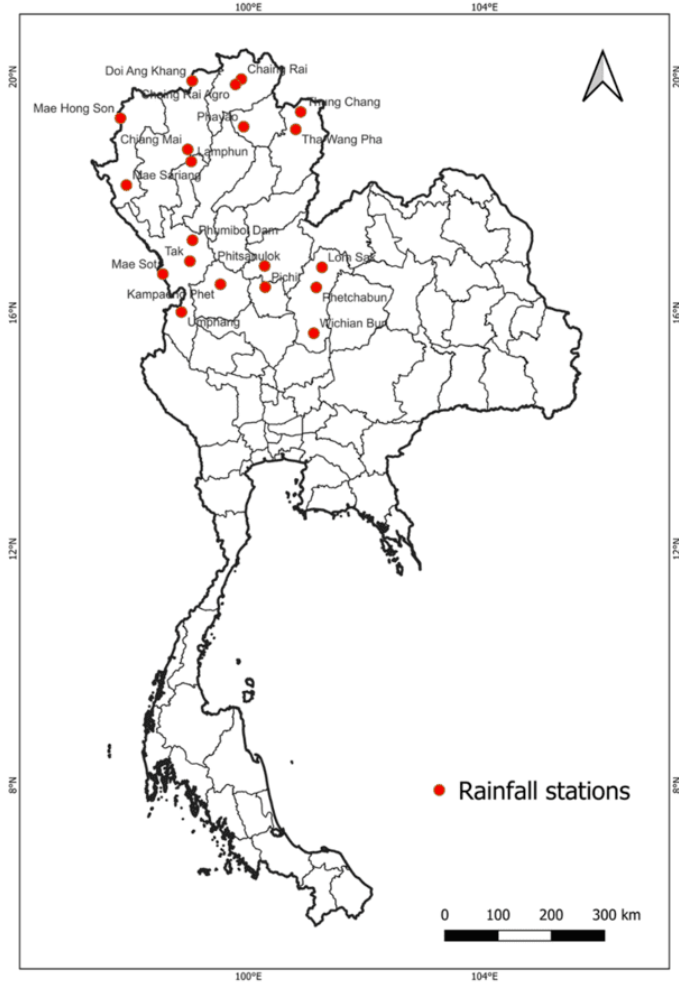


Fig. 1 Thailand's northern region, TMD rainfall stations

1.2. Data Analysis

The dataset under analysis encompasses 10957 days from 1993 to 2022, this daily rainfall data were collected across 31 TMD stations from northern Thailand (Fig. 1). Fig. 2 shows the rainfall stations exhibit varying degrees of missing data, as quantified by their respective missing percentages. Several key observations emerge from the analysis. Overall, the missing percentage varies from approximately 5 to 54%. Notably, stations such as "Doi Ang Khang" and "Mae Jo Agro" exhibit exceptionally high missing percentages, amounting to approximately 53.47% and 50.45% respectively. This substantial absence of data underscores the challenges associated with these stations and suggests caution in interpreting trends or patterns from

their records. Stations like "Thern" and "Sukhothai" also demonstrate relatively elevated missing percentages, at around 48.38% and 32.75%, respectively, warranting careful consideration when utilizing their data. On the other end of the spectrum, some stations present more complete datasets, with "Phrae" and "Nan" showcasing lower missing percentages at approximately 7.50% and 5.05%, respectively. This relative completeness enhances the reliability of their records for potential hydrological analyses and assessments.

Moreover, the dataset emphasizes the necessity of addressing missing data to ensure the accuracy and robustness of any hydrological investigations or modeling exercises. The presence of missing data can introduce biases and uncertainties in derived conclusions. Therefore, appropriate imputation methods or statistical techniques should be contemplated to mitigate these potential sources of error and to facilitate an additional detailed comprehension of the hydrological dynamics of the region. The collected meteorological data were subjected to statistical analysis. The dataset provides insight into the daily rainfall patterns across the region. The statistics encompass the mean, standard deviation, minimum, maximum, skewness, CV (coefficient of variation), and kurtosis for each station's rainfall records. Table 1 presents the descriptive and statistical analysis of rainfall data. The mean daily rainfall values, ranging from approximately 3.00 mm to 6.36 mm, offer insights into the average rainfall experienced at each station over the study period. Stations like "Doi Ang Khang" and "Mae Jo Agro" exhibit notably higher mean values, indicating that these locations receive consistently greater daily rainfall amounts than other stations. This could imply a more favorable environment for agricultural activities that rely on sufficient water supply. The standard deviation (SD) values, from around 8.37 mm to 13.46 mm, shed light on the dispersion or variability of daily rainfall around the mean. Stations with larger SD values, such as "Doi Ang Khang" and "Mae Jo Agro," suggest a higher degree of volatility in their rainfall patterns. This could indicate a greater likelihood of rapid shifts between periods of intense rainfall and dry spells, which has implications for water management strategies and infrastructure planning. The range of rainfall, represented by the difference between maximum and minimum values, demonstrates the extent of variability within each station's data. Stations with more comprehensive ranges, like "Thung Chang" and "Chaing Rai Agro," exhibit significant differences between the lowest and highest recorded rainfall amounts. In contrast, narrower ranges observed in stations like "Tha Wang Pha" and "Lom Sak" point towards more stable rainfall conditions with less extreme fluctuations. The positive skewness values observed in most stations' data distributions signify that the distribution of daily rainfall is skewed towards higher values. It suggests that while most days experience relatively moderate rainfall amounts, occasional days are characterized by intense and heavy rainfall events. These infrequent but

substantial rainfalls can significantly influence hydrological processes. The coefficient of variation (CV) values, ranging from approximately 209.90% to 325.47%, provide insights into the relative variability of daily rainfall about the mean. Stations with higher CV values, such as "Phumibol Dam" and "Uttaradit," exhibit substantial relative fluctuations in their daily rainfall amounts. This indicates that these locations experience a broad spectrum of relative changes in rainfall, which can affect water resource management and flood risk assessment. The positive kurtosis values observed, particularly in stations like "Phrae" and "Bhumibol Dam," indicate that daily rainfall peaks and has heavier tails than a normal distribution. It suggests an increased likelihood of extreme rainfall events, with a higher probability of observing intense and rare rainfall occurrences. The potential for such extreme events can have significant implications for infrastructure design and flood mitigation strategies. It can be concluded that the detailed examination of the provided statistical parameters underscores the intricate nature of daily rainfall patterns across the meteorological stations. These insights contribute to a comprehensive understanding of the hydrological characteristics of the region, aiding hydrologists, water resource managers, and policymakers in making informed decisions on water management, flood preparedness, and infrastructure planning

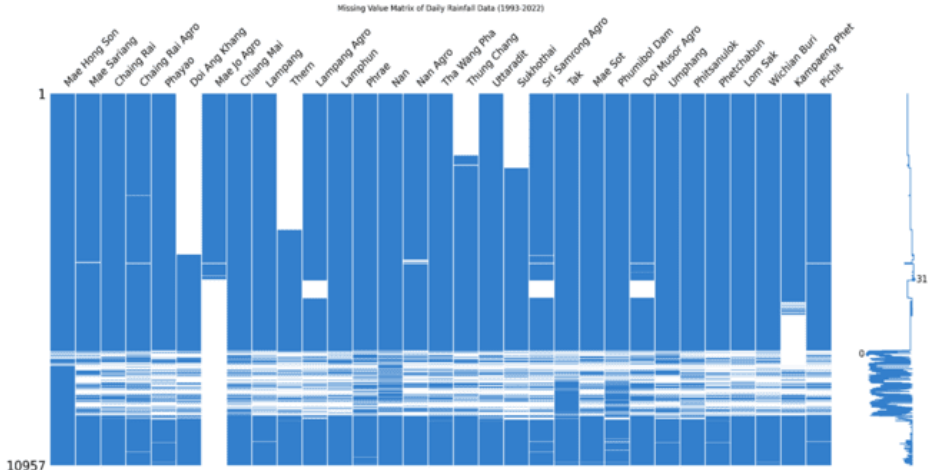


Fig. 2 Missing Daily Rainfall Data Matrix for all TMD stations in Northern Thailand

Table 1 Descriptive and Statistical Analysis summary of all TMD stations in Northern Thailand.

Rainfall Stations	Mean	SD	Min	Max	Skewness	CV	Kurtosis	% of Missing
Mae Hong Son	3.91	9.55	0	128.00	4.38	244.28	.2751	9.63
Mae Sariang	3.64	8.37	0	135.50	4.05	229.96	.2441	10.77
Chaing Rai	5.27	12.65	0	217.30	4.24	239.92	.2795	9.92
Chaing Rai Agro	5.01	11.99	0	147.40	4.04	239.24	.2239	10.33
Phayao	3.68	10.33	0	154.30	4.98	280.57	.3467	12.13
Doi Ang Khang	6.36	13.46	0	234.40	4.24	211.80	.3139	53.47
Mae Jo Agro	3.00	8.95	0	126.90	5.42	297.76	.4090	50.45
Chiang Mai	3.54	9.68	0	144.40	5.06	273.08	.3706	9.56
Lampang	3.46	9.64	0	191.20	5.21	278.99	.4288	11.60
Thern	4.19	10.98	0	156.50	4.59	262.07	.2913	48.38
Lampang Agro	3.82	9.82	0	123.10	4.14	256.98	.2257	16.74
Lamphun	3.29	9.38	0	156.00	4.73	284.80	.3111	9.77
Phrae	3.49	10.23	0	218.20	5.81	293.25	.5389	7.50
Nan	3.68	10.48	0	143.50	4.87	284.64	.3231	5.05
Nan Agro	3.95	10.68	0	161.30	4.72	270.26	.3190	12.48
Tha Wang Pha	4.51	11.85	0	141.80	4.56	262.59	.2806	9.96
Thung Chang	5.19	12.97	0	259.00	4.98	249.73	.4255	27.95
Uttaradit	4.21	12.01	0	263.70	5.53	285.18	.5246	11.23
Sukhothai	4.28	12.00	0	208.80	5.43	280.64	.4486	32.75
Sri Samrong Agro	3.87	10.98	0	176.70	5.16	284.00	.3933	17.65
Tak	3.21	1	0	17	6.	318.	52	5.93

		0.23		1.50	04	54	.87	
Mae Sot	4.69	1 1.55	0	20 7.40	5. 04	246. 57	41 .53	10.60
Phumibol Dam	3.23	1 0.50	0	24 7.10	6. 05	325. 47	60 .48	7.56
Doi Musor Agro	4.76	1 0.98	0	26 1.10	5. 16	230. 43	53 .75	14.50
Umphang	4.55	9. 56	0	12 4.70	3. 76	209. 90	20 .01	9.39
Phitsanulok	4.09	1 1.40	0	16 7.10	4. 64	278. 62	28 .93	9.96
Phetchabun	3.60	9. 61	0	17 0.40	5. 03	267. 02	39 .47	12.12
Lom Sak	3.24	9. 04	0	13 2.30	4. 93	279. 19	33 .22	11.96
Wichian Buri	4.01	1 1.54	0	24 3.80	5. 48	287. 64	49 .85	11.51
Kampaeng Phet	4.04	1 0.89	0	24 8.90	5. 36	269. 56	54 .16	23.56
Pichit	3.77	1 0.64	0	14 0.80	4. 59	282. 53	27 .88	11.24

1.3. Correlation analysis between all rainfall stations

Pearson's correlation coefficient is employed to evaluate the nature and magnitude of a linear relationship among two continuous variables. It is represented by "r," which ranges between -1 and +1. Values nearer to +1 argue a robust positive linear relationship, values which is nearer to -1 signify a sturdy negative linear relationship and ranges near 0 suggest slight to no linear relation [26, 27].

Fig. 3 provides a correlation matrix that displays Pearson's correlation coefficients between different rainfall stations. Each cell in the matrix represents the correlation between the two stations. Firstly, it is evident that there is a considerable degree of spatial variability in rainfall patterns among the studied stations. It is highlighted by the range of correlation coefficients, varying from weak positive to strong positive correlations. These correlations signify the extent to which rainfall patterns at one station relate to those at another. Mae Hong Son and Mae Sariang exhibit a moderate positive correlation (0.28), suggesting that when rainfall increases in Mae Hong Son, a similar increase tends to occur in Mae Sariang. Though not exceptionally strong, this connection implies some degree of synchronization in their rainfall patterns. Similarly, Mae Hong Son shares a weak positive correlation (0.25) with Chiang Rai, indicating a slight tendency for their rainfall to coincide. This association is reinforced by another weak positive correlation (0.27) between Mae Hong Son and Chiang Rai Agro, underscoring a mild parallelism in their rainfall variations.

Furthermore, the moderate positive correlation (0.29) between Mae Hong Son and Phayao implies a notable yet not overwhelming link between their rainfall trends. When Mae Hong Son experiences higher or lower rainfall, Phayao follows suit. Similarly, Doi Ang Khang and Mae Hong Son exhibit a moderate positive correlation (0.32), implying that changes in rainfall are moderately aligned between these two stations. Turning to Mae Jo Agro, a moderate positive correlation (0.32) indicates that it shares a propensity for corresponding changes in rainfall with Mae Hong Son. This connection could be attributed to their geographical proximity. Chiang Mai, situated further away, still maintains a moderate positive correlation (0.30) with Mae Hong Son, suggesting that their rainfall patterns tend to move in tandem despite the distance. The relationship extends to other stations, like Lampang, with its moderate positive correlation (0.29) with Mae Hong Son. Though not perfectly synchronized, their rainfall patterns do share some similarities. The correlation coefficient between Mae Hong Son and Thern is weaker (0.27), indicating a milder connection between their rainfall variations. The correlation between Mae Hong Son and Lampang Agro, on the other hand, is moderately positive (0.30), suggesting that their rainfall trends are moderately correlated. Most importantly, the correlation between Tha Wang Pha and Nan is 0.54, indicating a moderate positive relationship in their daily rainfall patterns. It suggests that these two stations experienced

similar weather conditions over the studied period. Next, the correlation between Tha Wang Pha and Nan Agro station's data is 0.63, signifying a strong positive relationship. Nan Agro appears to have a stronger connection to the rainfall patterns observed at the Tha Wang Pha station. This could imply that the agricultural area (Nan Agro) might be significantly influenced by the rainfall observed in the nearby Tha Wang Pha station, which is valuable information for crop planning and agricultural water management. The correlation value of 0.27 between Tha Wang Pha and Phrae station's data indicates a relatively weak positive relationship. These two stations share some commonality in their rainfall patterns, but the correlation suggests that different meteorological conditions influence them somewhat.

Similarly, the correlation between Tha Wang Pha and Thung Chang station's data is 0.57, suggesting a strong positive relationship. It implies that the rainfall patterns observed at these two stations are somewhat related but not strongly so. Understanding these relationships can aid in assessing regional climate variations and their impact on water resources. Lastly, the correlation between Tha Wang Pha and Uttaradit is 0.25, demonstrating a weak positive relationship. While these stations are not highly correlated, the positive association suggests that they share some similarities in their rainfall patterns, which could be attributed to their geographical proximity. As depicted in Fig.s 1 and 2, the daily rainfall dataset exhibits instances of missing observations across all stations. The interrelations among these stations, as gauged by the correlation coefficients, reflect a commendable degree of association. Specifically, the correlation between Tha Wang Pha station's rainfall data and its neighboring stations (Nan, Nan Agro, Phrae, and Uttaradit). Turning our attention to Table 1, we encounter the percentage representation of absent data for distinct rainfall stations. The Thung Chang station has the most substantial prevalence of missing data, accounting for 27.95% of its dataset. Nan Agro follows suit sequentially with a missing data percentage of 12.74%, implying a relatively more comprehensive dataset. Uttaradit and Tha Wang Pha exhibit analogous missing data percentages, approximating 11.23% and 9.95%, respectively, indicative of a moderate degree of data absence. Finally, the Nan rainfall station has the lowest missing data percentage of 5.05%, implying a somewhat lower proportion of data unavailability among these five stations. These percentages concerning absent data afford valuable insights into the plausible limitations that could arise while analyzing or construing the rainfall trends for these stations. Drawing from the criteria of geographical proximity to neighboring stations and the extent of data gaps, these stations are initially selected for preliminary estimation endeavors. Based on the correlation between neighboring stations and the percentage of missingness of daily rainfall data, these five stations are selected firstly to impute missing values by K-Nearest Neighbors (KNN).



Fig. 3 Pearson's Correlation Heatmap between all TMD rainfall stations in Northern Thailand

1.4. Data Imputation using k-Nearest Neighbors (kNN) for Missing Rainfall Data

This section presents the methodology employed to establish missing values in the previously mentioned rainfall station's statistics. Missing values in the dataset can compromise its integrity and limit its usability for further analysis. A KNN imputation approach is adopted to address this issue. A KNN imputer is utilized to handle the missing values. KNN is an ML algorithm that leverages similarity measures to attribute missing values constructed on the patterns perceived in the dataset. Here, the imputer is initialized with a parameter "n_neighbors," which defines the number of neighboring data points to consider during imputation. The imputation process involves identifying missing values and replacing them with estimated values derived from the neighboring data points. In this case, the KNN imputer uses the available rainfall data from other stations to predict and impute missing values.

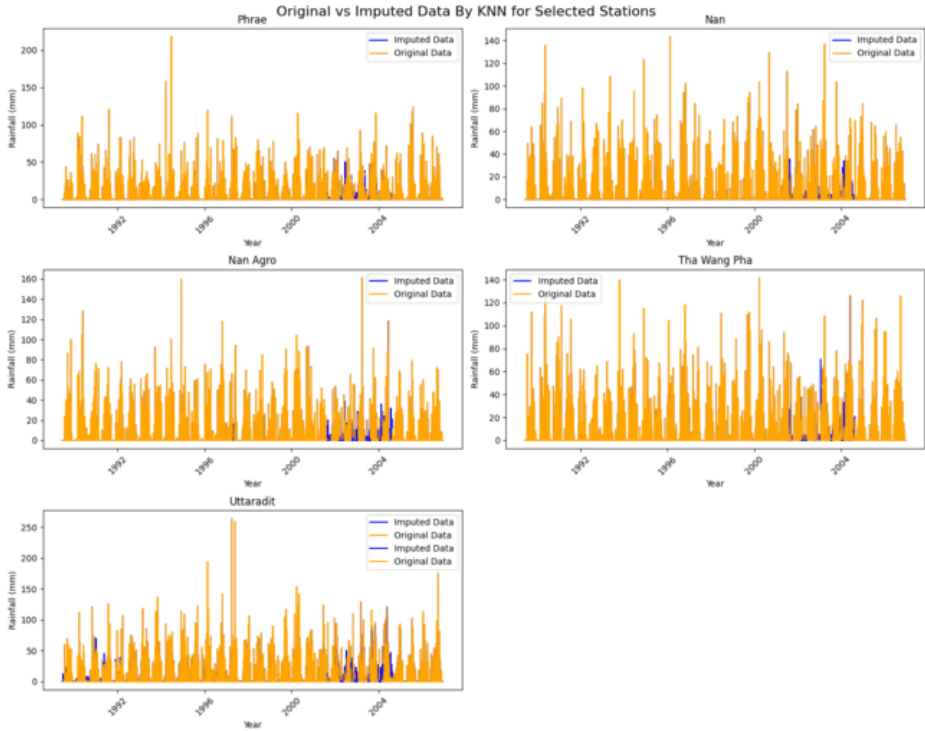
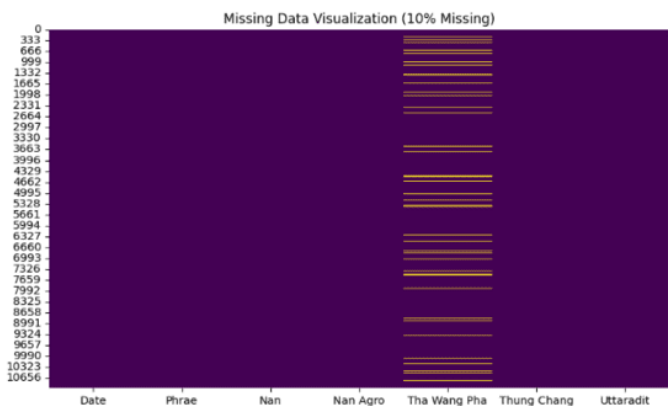
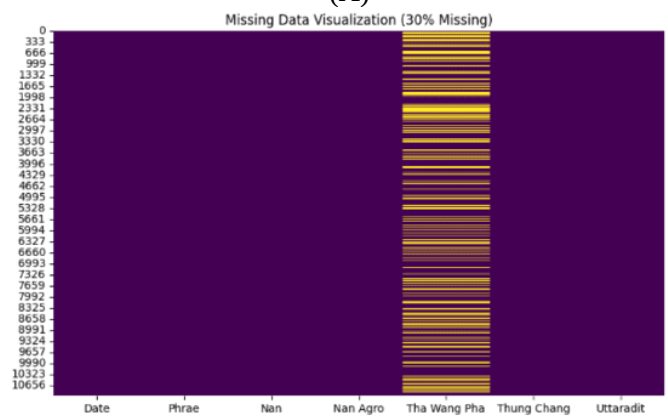


Fig. 3 Original and Imputed data by kNN at all selected stations

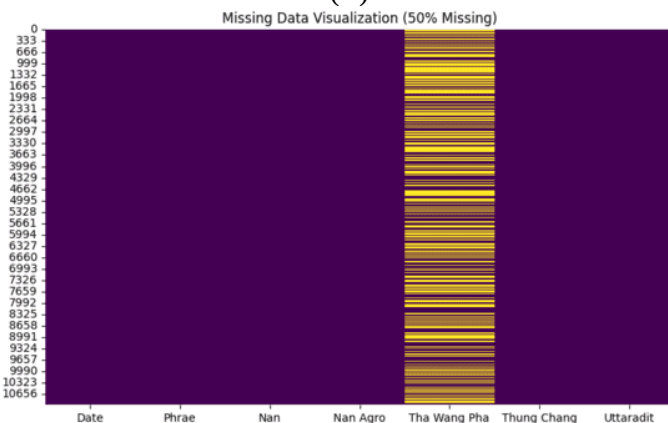
After using the KNN method to complete the dataset (Fig. 3), we have successfully obtained comprehensive datasets for five distinct rainfall measurement stations. This has resulted in establishing a complete reference dataset, which is now ready for validation. After imputing missing values at all stations by KNN and based on the correlation of Tha Wang Pha station with other neighboring stations was chosen as a targeted station. The 10%, 30%, and 50% of daily data from Tha Wang Pha station dataset was randomly deleted. Fig. 4 shows the random deletion of the daily dataset. The next phase of this research involves thoroughly evaluating various methods for estimating daily rainfall data. These methods include AA, NIPLAS, MLR, M5-MT, and LSTM-RNN. To ensure a comprehensive evaluation, we have intentionally introduced random data gaps within the target rainfall station based on Figs 1 and 2 from lowest to highest missing percentage. By adopting this approach, we aim to assess how effectively these estimation techniques perform when faced with data gaps or incomplete information. This assessment will provide significant comprehension of the resilience and efficacy of the different methods under limited data conditions, helping us understand their relative performance and potential suitability in situations where data may be lacking.



(A)



(B)



(C)

Fig. 4 Randomly deleted dataset from Tha Wang Pha station (a) 10% randomly deletion of daily data, (b) 30% randomly deletion of daily data, (c) 50% randomly deletion of daily data.

1.5. Methods for Estimation of Missing Data

Estimating missing data in hydrological modeling and weather forecasting is very crucial. Many techniques for estimating and recreating missing data developed. These methods are classified as function-fitting techniques and statistical or empirical methods. Most of these approaches utilize observations collected from nearby stations to fill in the missing values. Choosing appropriate strategies for the interpretation of missing rainfall data improves the accuracy of hydrological models [3]. This study examines the capacity of various conventional methods to calculate missing rainfall data over Thailand and identify the most suitable method. The comprehensive approach for managing missing data is depicted in Fig. 5.

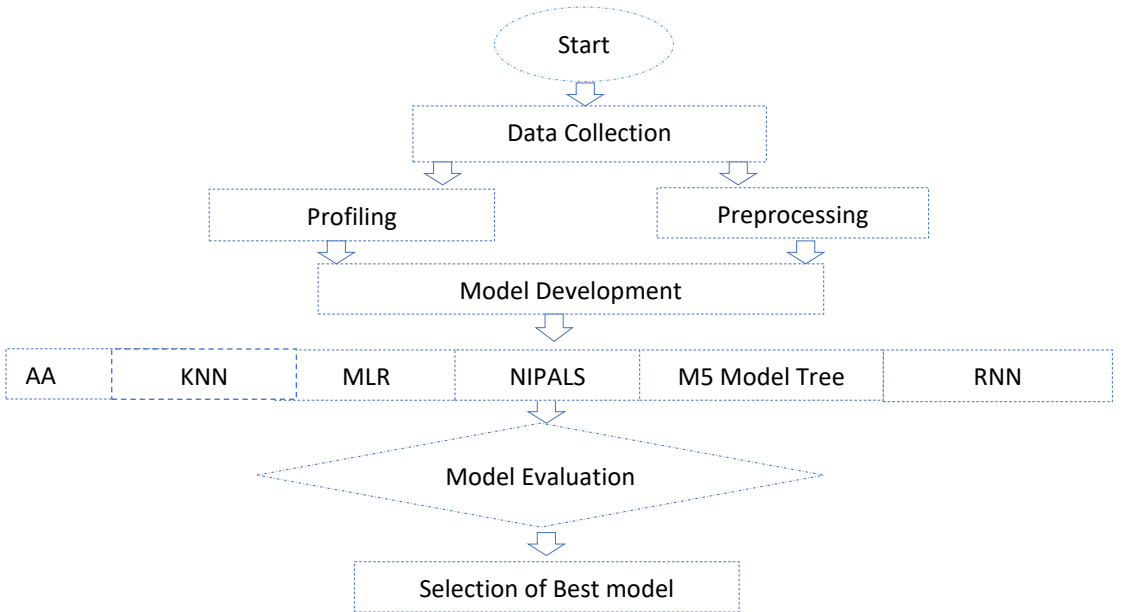


Fig. 5 Schematic diagram for methodology overview to handling missing data

1.5.1. Arithmetic Averaging

The simplest method, which gives the missing data by computing the arithmetic mean of the data correlating to the nearest weather stations, is frequently used in meteorology to fill in missing weather data, as shown in equation 9.

$$Po = \frac{\sum_{i=1}^n Pi}{N} \quad (1)$$

Missing data estimation is represented as P_o , the similar value of an identical parameter of the adjoining meteorological stations P_i , and point of interest which from the nearest station number is denoted as N . AA technique is suitable when the gauges are evenly distributed across the region, and individual gauge readings do not exhibit significant deviations from the mean [28].

1.5.2. Multiple linear regression analysis

MLR is a statistical technique used to assess the relationship between the independent (two or more) and dependent variables. MLR identifies the optimal combination of independent variables' weights to predict the dependent variable accurately. Reference [29] emphasized the numerous benefits of using this technique for missing data prediction and data interpolation. Estimating of missing data (P_o) is determined by equation 10.

$$P_o = a_o + \sum_{i=1}^n (a_i \times P_i) \quad (2)$$

where $a_i, a_1, a_2, \dots, a_n$ are the coefficients of regression.

1.5.3. Non-linear iterative partial least squares (NIPALS) algorithm

The NIPALS algorithm within NILES was initially introduced by (Wold 1966). The process involves iterative principal component analysis on data sets containing missing values. The primary aim is to determine the slope of the least-squares line passing through the origin of the observed data points.

The eigenvalues are determined by adjusting the NIPALS components, and a similar approach is applied to estimate the missing data. The algorithm's convergence rate is directly influenced by the proportion of missing data [28].

1.5.4. M5 Model Trees

The M5-MT model, a variant of the model developed [30], substitutes linear functions for discrete class labels at the leaves of the model[31]. The M5 model operates in a top-down manner, employing a divide-and-conquer strategy(Witten and Frank 2002). The SDR serves as the criterion for division formula (12):

$$SDR = SD(t) - \sum_{i=0}^n \frac{|t_i|}{|t|} \times SD(t_i) \quad (3)$$

"t" represents the set of samples that reached the node, t_i is the sample subset with the i th possible outcome, and the standard deviation is denoted as "SD". This approach reduces the standard deviation of the child nodes. As stated by Quinlan (1992), the model selects the final split to minimize the reduction in predicted error. However, there's a risk of the model becoming excessively complex due to the overfitting of test data.

1.5.5. Recurrent neural networks (RNNs)

RNNs, including LSTM and gated recurrent units, [32] have demonstrated the ability to generate deep hierarchical features and attain state-of-the-art performance across various real-world applications involving multivariate time series data. Within time series data, they can effectively recognize robust long-range relationships. Several contemporary approaches for addressing missing values in RNNs involve aligning diverse sampling frequencies, concatenating absent entries, and integrating a time-based decay function. In time series datasets, like meteorological observations, the occurrence of missing values is a significant concern. Data completeness is necessary for time series analysis [33].

RNN is one of the most extensively employed deep learning techniques for time series analysis. RNN implies that each sample is dependent on the one preceding it. In some studies, RNNs provide excellent performance for predicting time series data, but the RNNs' ability to learn long-range dependencies is limited by the vanishing gradient problem. In many real-world applications, it is essential to consider both short- and long-term dependencies. A variant of RNN known as LSTM addresses the challenge of vanishing gradients by enabling the retention or forgetting of information for each state. [34]. RNN-LSTM, RNN, and other algorithms with RNN will be tested to estimate missing daily rainfall data.

III. RESULTS AND DISCUSSIONS

In this study, a comprehensive 30-year daily rainfall dataset from the Tha Wang Pha (Target station) and Nan, Nan Agro, Phrae, Thung Chang, and Uttaradit TMD stations (neighboring stations) was utilized. To emulate varying levels of data incompleteness, random data deletion was applied, resulting in three distinct test datasets: Set-1, comprising 10% of daily data randomly deleted; Set-2, with 30% of daily data deleted; and Set-3, exhibiting a more substantial 50% of daily data deletion. The decision to split the data in this manner was predicated on the observed overall data missingness pattern across multiple TMD stations in Thailand. Before the random deletion of data, a thorough review was conducted to ensure the representativeness of the missing values. Subsequently, the imputation techniques were employed to estimate and replace the missing values in all three sets.

Fig. 6 shows the radar map and evaluation metrics results of imputation methods for Set-1, encompassing 10% of missing daily rainfall data. AA demonstrated strong performance with an R^2 value of 0.93, indicating that the imputed values explained approximately 93% of the variance in the observed data. The MSE of 8.54, MAE of 0.36, and RMSE of 2.92 collectively suggest highly accurate imputations with relatively small errors. NIPALS showed reasonable performance but exhibited a lower R^2 value of 0.85 than AA. The MSE of 18.93, MAE of 0.65, and RMSE of 4.35 indicated a more significant imputation error margin. M5-MT demonstrated respectable performance, with an R^2 value of 0.89 and moderately higher MSE, MAE, and RMSE than MLR and AA. KNN showed performance closely aligned with AA, with an R^2 value of 0.93 and similar MSE, MAE, and RMSE values. Most importantly, MLR performed impressively well, with an R^2 value of 0.94, signifying the explanatory solid power of the imputed values. The MSE of 7.5, MAE of 0.35, and RMSE of 2.74 underlined the high accuracy and precision of the imputations. LSTM-RNN yielded results like MLR, with an R^2 value of 0.93 and similar MSE, MAE, and RMSE values, suggesting success in capturing underlying data patterns.

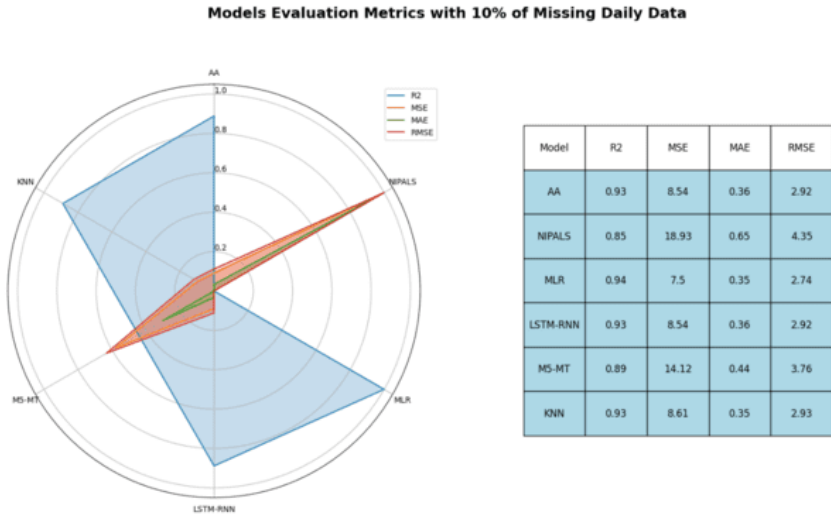


Fig. 6 Model's performance with 10% of missing daily rainfall dataset (SET-1)

Fig. 7 shows the radar map and evaluation metrics results of imputation methods for Set-2, encompassing 30% of missing daily rainfall data. The AA method exhibited an R^2 value of 0.82, indicating that the imputed values explained approximately 82% of the variance in the observed data. However, it's notable to perceive that the MSE of 23, MAE of 1.04, and RMSE of 4.8 suggest a relatively higher error level in the imputations, likely due to the increased complexity of handling 30% missing data. NIPALS, in this case, demonstrated lower performance, reflected by an R^2 value of 0.59, indicating that the imputed values captured a smaller proportion of variance in the observed data. The MSE of 52.22, MAE of 1.91, and RMSE of 7.23 collectively indicate a substantial error margin in the imputations, making this method less accurate for the task at hand. MLR showcased a relatively more robust performance, as evidenced by an R^2 value of 0.85, signifying that the imputed values accounted for 85% of the variance in the observed data. The MSE of 18.96 and MAE of 0.98 imply a lower error level, although the RMSE of 4.35 suggests some degree of variability. LSTM-RNN yielded an R^2 value of 0.82, closely aligned with AA, implying that the imputed values explain approximately 82% of the variance in the observed data. Like AA, the MSE of 23, MAE of 1.04, and RMSE of 4.8 indicate challenges in handling the increased data gaps in Set-2. M5-MT exhibited a lower R^2 value of 0.65, implying that the imputed values captured a modest proportion of variance. The MSE of 45.18, MAE of 1.42, and RMSE of 6.72 suggest a relatively higher imputation error rate. KNN displayed performance closely resembling AA, with an R^2 value of 0.81. The MSE of 23.78 and MAE of 1.09, accompanied by an RMSE of 4.88, indicate moderate imputation accuracy. Among the imputation methods assessed for Set-2, MLR, and LSTM-RNN emerge as the most effective choice. It achieved the highest R^2 value and demonstrated lower error metrics (MSE,

MAE, and RMSE) than other methods. Despite the increased data gaps in Set-2, MLR, and LSTM-RNN proved to be a robust methodology for attributing missing data of daily rainfall, providing accurate estimates of the missing values within the constraints of this scenario.

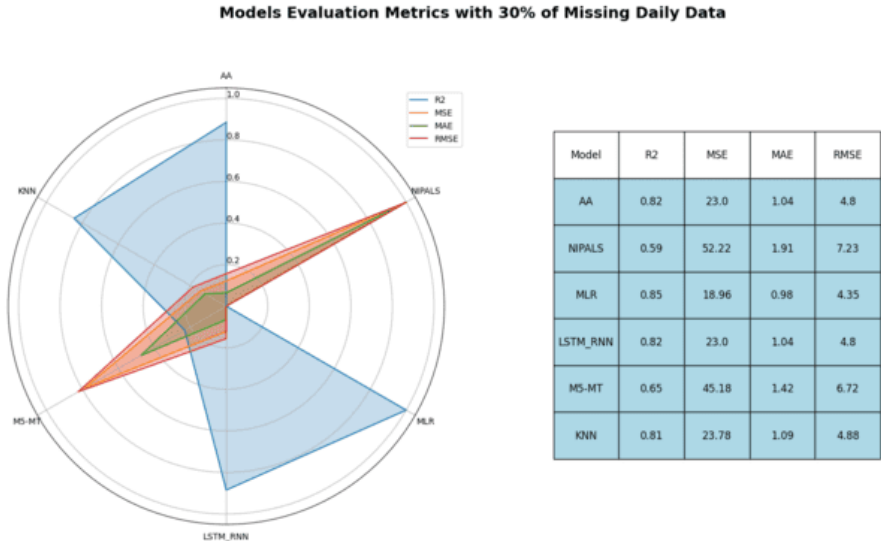


Fig. 7 Model's performance with 30% of missing daily rainfall dataset (SET-2)

Fig. 8 shows the radar map and evaluation metrics results of imputation methods for Set-2, encompassing 30% of missing daily rainfall data. The AA method demonstrated an R^2 (0.68), indicating that the imputed values accounted for approximately 68% of the variance in the observed data. However, it's essential to note that the MSE of 40.39, MAE of 1.79, and RMSE of 6.36 suggest a considerable degree of error in the imputations due to the substantial extent of missing data. In this context, NIPALS exhibited notably lower performance with an R^2 value of 0.24, implying a limited capability to capture variance in observed statistics using imputed values. The MSE of 97.17, MAE of 3.28, and RMSE of 9.86 collectively underscore significant imputation errors, making it less suitable for this challenging scenario. MLR showcased a relatively better performance, as reflected by an R^2 (0.74), indicating that the imputed values explained 74% of the variance in the observed data. The MSE of 33.06 and MAE of 1.69, along with an RMSE of 5.75, suggest a moderate level of imputation error, which is relatively lower given the substantial data gaps. LSTM-RNN demonstrated a commendable R^2 (0.75), indicating that the imputed values effectively capture approximately 75% of the variance in the observed data. The MSE of 32.06, MAE of 1.51, and RMSE of 5.15 collectively indicate a relatively low error level in the imputations, considering the challenging nature of Set-3 with 50% missing data. M5-MT exhibited an R^2 (0.36), indicating that the imputed values captured only a portion of the variance. The MSE of 82.03 and MAE of 2.4, along with an RMSE of 9.06, point to a relatively higher

level of imputation error. KNN displayed performance like AA, with an R^2 (0.68). The MSE of 40.58 and MAE of 1.84, accompanied by an RMSE of 6.37, indicate moderate imputation accuracy in the context of Set-3's challenging data gaps. Considering the demanding scenario of Set-3 with 50% missing data, LSTM-RNN stands out as the most effective imputation method.

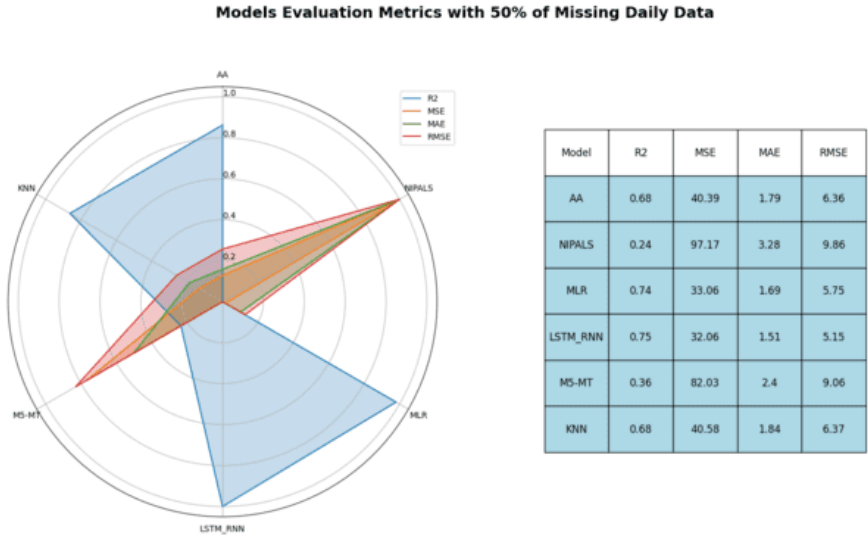


Fig. 8 Model's performance with 50% of missing daily rainfall dataset (SET-3)

IV. CONCLUSION

In conclusion, this study employed various estimation methods to estimate missing values in three distinct datasets with different levels of data incompleteness: Set-1 (10% missing data), Set-2 (30% missing data), and Set-3 (50% missing data). For Set-1, MLR demonstrated strong performance, outperforming other methods with a high R^2 (0.94) and relatively small error metrics. MLR also performed impressively well, with an R^2 (0.94), indicating strong explanatory power. In Set-2, MLR and LSTM-RNN emerged as the most effective choices, achieving the highest R^2 and demonstrating relatively lower error metrics than other methods. Despite the increased complexity of handling 30% of missing data, MLR and LSTM-RNN proved to be robust approaches for accurate imputations. For the challenging Set-3, with 50% missing data, LSTM-RNN stood out as the most effective imputation method, showcasing the highest R^2 and relatively low error metrics. Its ability to capture temporal dependencies and patterns in sequential data proved beneficial in handling the substantial data gaps in this scenario. While each imputation method showed varying performance across different datasets, MLR consistently performed well. LSTM-RNN demonstrated effectiveness in handling increased data gaps, especially in the challenging scenario of Set-3 with 50% missing data. The choice of the most

accurate estimation method may depend on the explicit characteristics of the dataset and the nature of the missing data. Based on this analysis, the LSTM-RNN method is preferred for estimating missing values across all TMD stations and variables.

ACKNOWLEDGMENT

The authors would like to express their gratitude to The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi, and the Center of Excellence on Energy Technology and Environment (CEE), Ministry of Higher Education, Science (MHESI), Research and Innovation and Department of Mathematics for the financial and technical support that allowed them to conduct this study. The authors also would like to thank you, Department of Mathematics, King Mongkut's University of Technology Thonburi, for providing us with this opportunity.

CONFLICTS OF INTEREST

The authors declare that the research was conducted without any commercial or financial relationships that could be construed as a potential conflict of interest.

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